"

lead4ward

IQ: INVESTIGATING THE QUESTIONS

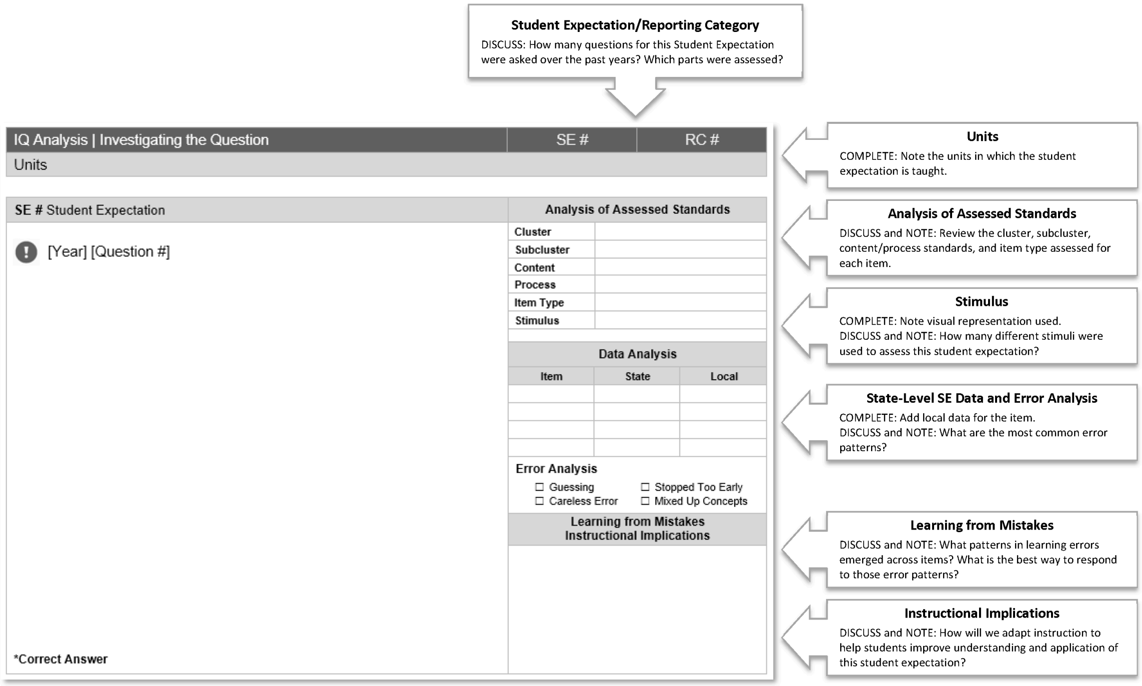
2019-2025 Released Tests

Aligned to the Standards

CONTENT BUILDER FOR THE PLC

Science
Grade 5

IQ Analysis | Investigating the Question – Released Tests User Guide



In conjunction with the IQ analysis tool, the lead4ward field guides can be a helpful resource for understanding error patterns and instructional implications.

Learn more

Properties of Matter

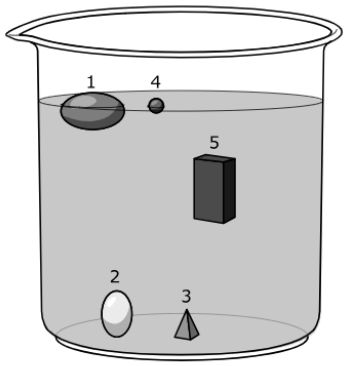
5.6 Matter and energy. The student knows that matter has measurable physical properties that determine how matter is identified, classified, changed, and used.

IQ Analysis | Investigating the Question	5.6(A)	RC 1

5.6(A) compare and contrast matter based on measurable, testable, or observable physical properties, including mass, magnetism, relative density (sinking and floating using water as a reference point), physical state (solid, liquid, gas), volume, solubility in water, and the ability to conduct or insulate thermal energy and electric energy

2025 – Q7

A student drops five objects into a container of water.



Which statement **BEST** classifies some of these objects?

Ⓐ Objects 2 and 5 are more dense than water, while Objects 1 and 4 are less dense than water.

Ⓑ Objects 1 and 4 are more dense than water, while Objects 2 and 3 have the same density as water.

Ⓒ Objects 2 and 3 are less dense than water, while Object 5 is more dense than water.

Ⓓ Objects 1 and 4 are less dense than water, while Object 2 is more dense than water.

***Correct Answer (D)**
^highly chosen incorrect answer (A)

Analysis of Assessed Standards

Cluster	Properties of Matter
Subcluster	Physical Properties
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A	^	
B		
C		
D*	64	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts
▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.6(A) compare and contrast matter based on measurable, testable, or observable physical properties, including mass, magnetism, relative density (sinking and floating using water as a reference point), physical state (solid, liquid, gas), volume, solubility in water, and the ability to conduct or insulate thermal energy and electric energy

2025 – Q16

A student groups the following materials together:

- copper wire
- brick
- wood
- rubber

Which property did the student **MOST LIKELY** use to group the materials?

Ⓐ Ability to float in water

Ⓑ Physical state

Ⓒ Flexibility

Ⓓ Electrical conductivity

***Correct Answer (B)**
^highly chosen incorrect answer (D)

Analysis of Assessed Standards

Cluster	Properties of Matter
Subcluster	Physical Properties
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A		
B*	41	
C		
D	^	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.6(A) compare and contrast matter based on measurable, testable, or observable physical properties, including mass, magnetism, relative density (sinking and floating using water as a reference point), physical state (solid, liquid, gas), volume, solubility in water, and the ability to conduct or insulate thermal energy and electric energy

OLD **5.5(A)** classify matter based on measurable, testable, and observable physical properties, including mass, magnetism, physical state (solid, liquid, and gas), relative density (sinking and floating using water as a reference point), solubility in water, and the ability to conduct or insulate thermal energy or electric energy

Analysis of Assessed Standards

2024 – Q3

This question has two parts. First, answer Part A. Then answer Part B.

Students perform an investigation to determine which of four materials will keep a hot drink warm for the longest period of time.

The students gather four different cups, each made of a different material. The cups are the same size. The teacher pours equal amounts of heated water into the cups.

The students record the temperature of the water in each cup every minute for 4 minutes. The results are shown in the table.

Material	Temperature (°C)				
	0 Minutes	1 Minute	2 Minutes	3 Minutes	4 Minutes
1	100	92	84	75	67
2	100	90	82	73	64
3	100	85	70	65	50
4	100	96	92	88	84

Part A
Which material will keep a hot drink warm for the longest period of time?

Ⓐ Material 1

Ⓑ Material 2

Ⓒ Material 3

Ⓓ Material 4

Part B
Which statement **BEST** supports the answer to Part A?

Ⓐ The material that will keep the drink warm longest insulates thermal energy inside the cup.

Ⓑ The material that will keep the drink warm longest conducts thermal energy into the cup.

Ⓒ The material that will keep the drink warm longest conducts thermal energy away from the cup.

Ⓓ The material that will keep the drink warm longest insulates thermal energy from entering the cup.

***Correct Answer (D, A)**

Cluster	Properties of Matter
Subcluster	Physical Properties
Content	Readiness
Process	
Item Type	Multipart (2 pts)
Stimulus	

Data Analysis

Item	State	Local
Full Credit	46	
No Credit	10	
Partial Credit	44	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

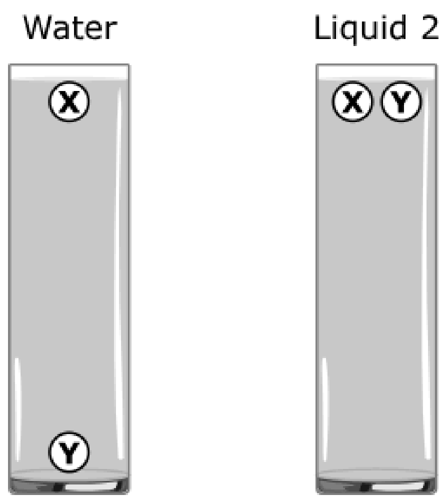
5.6(A) compare and contrast matter based on measurable, testable, or observable physical properties, including mass, magnetism, relative density (sinking and floating using water as a reference point), physical state (solid, liquid, gas), volume, solubility in water, and the ability to conduct or insulate thermal energy and electric energy

OLD **5.5(A)** classify matter based on measurable, testable, and observable physical properties, including mass, magnetism, physical state (solid, liquid, and gas), relative density (sinking and floating using water as a reference point), solubility in water, and the ability to conduct or insulate thermal energy or electric energy

Analysis of Assessed Standards

2024 – Q25

A student conducted an investigation about density. The materials used were Water, Liquid 2, Object X, and Object Y. The results of the investigation are shown.



Which option shows the correct order of the materials from lowest to highest density?

Ⓐ Object X, Object Y, Water, Liquid 2

Ⓑ Water, Liquid 2, Object X, Object Y

Ⓒ Water, Object X, Liquid 2, Object Y

Ⓓ Object X, Water, Object Y, Liquid 2

***Correct Answer (D)**
^highly chosen incorrect answer (B)

Cluster	Properties of Matter
Subcluster	Physical Properties
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A		
B	^	
C		
D*	30	

Error Analysis
- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.6(A) compare and contrast matter based on measurable, testable, or observable physical properties, including mass, magnetism, relative density (sinking and floating using water as a reference point), physical state (solid, liquid, gas), volume, solubility in water, and the ability to conduct or insulate thermal energy and electric energy

OLD **5.5(A)** classify matter based on measurable, testable, and observable physical properties, including mass, magnetism, physical state (solid, liquid, and gas), relative density (sinking and floating using water as a reference point), solubility in water, and the ability to conduct or insulate thermal energy or electric energy

Analysis of Assessed Standards

2023 – Q3

Students poured water and cooking oil into a beaker. The students then added a penny, a plastic paper clip, and a cork to the beaker and recorded their observations.

Observations:

- The water was on the bottom, and the oil was on top of the water.
- The penny dropped to the bottom of the beaker.
- The plastic paper clip floated on the water but under the oil.
- The cork floated on top of the oil.

Which list puts the items in order from most dense to least dense?

Ⓐ Cork, oil, plastic paper clip, water, penny

Ⓑ Water, penny, oil, plastic paper clip, cork

Ⓒ Penny, water, plastic paper clip, oil, cork

Ⓓ Oil, penny, cork, plastic paper clip, water

***Correct Answer (C)**

Cluster	Properties of Matter
Subcluster	Physical Properties
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A	22	
B	14	
C*	56	
D	9	

Error Analysis

- ☐ Guessing
- ☐ Mixed Up Concepts
- ☐ Careless Error
- ☐ Stopped Too Early

Learning from Mistakes Instructional Implications

5.6(A) compare and contrast matter based on measurable, testable, or observable physical properties, including mass, magnetism, relative density (sinking and floating using water as a reference point), physical state (solid, liquid, gas), volume, solubility in water, and the ability to conduct or insulate thermal energy and electric energy

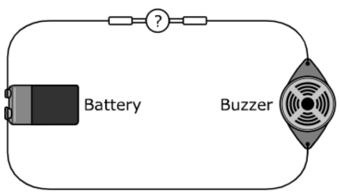
OLD **5.5(A)** classify matter based on measurable, testable, and observable physical properties, including mass, magnetism, physical state (solid, liquid, and gas), relative density (sinking and floating using water as a reference point), solubility in water, and the ability to conduct or insulate thermal energy or electric energy

Analysis of Assessed Standards

2023 – Q24

Students use a circuit to investigate the properties of six objects. The buzzer makes a sound if an object completes the circuit. The circuit and observations are shown.

Circuit



Observations

Object	Buzzer sounds?
1. Eraser	No
2. Toothpick	No
3. Paper clip	Yes
4. Cotton ball	No
5. Safety pin	Yes
6. Aluminum foil	Yes

Which objects are electrical insulators?

Ⓐ Eraser, toothpick, cotton ball

Ⓒ Toothpick, paper clip, safety pin

Ⓑ Eraser, paper clip, safety pin

Ⓓ Paper clip, safety pin, aluminum foil

***Correct Answer (A)**

Cluster	Properties of Matter
Subcluster	Physical Properties
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A*	54	
B	2	
C	3	
D	40	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.6(A) compare and contrast matter based on measurable, testable, or observable physical properties, including mass, magnetism, relative density (sinking and floating using water as a reference point), physical state (solid, liquid, gas), volume, solubility in water, and the ability to conduct or insulate thermal energy and electric energy

OLD **5.5(A)** classify matter based on measurable, testable, and observable physical properties, including mass, magnetism, physical state (solid, liquid, and gas), relative density (sinking and floating using water as a reference point), solubility in water, and the ability to conduct or insulate thermal energy or electric energy

Analysis of Assessed Standards

Cluster	Properties of Matter
Subcluster	Physical Properties
Content	Readiness
Process	5.2(D)
Stimulus	

2022 – Q4

4 A student places objects in a bucket of water to determine if they will float.

Object
Toothpick
Plastic paper clip
Penny
Cork
Metal spoon
Vegetable oil

Which set of items is less dense than water?

F Plastic paper clip, penny, and cork

G Toothpick, metal spoon, and plastic paper clip

H Metal spoon, vegetable oil, and penny

J Toothpick, cork, and vegetable oil

***Correct Answer (J)**

Data Analysis

Item	State	Local
F	13	
G	7	
H	8	
J*	73	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.6(A) compare and contrast matter based on measurable, testable, or observable physical properties, including mass, magnetism, relative density (sinking and floating using water as a reference point), physical state (solid, liquid, gas), volume, solubility in water, and the ability to conduct or insulate thermal energy and electric energy

OLD **5.5(A)** classify matter based on measurable, testable, and observable physical properties, including mass, magnetism, physical state (solid, liquid, and gas), relative density (sinking and floating using water as a reference point), solubility in water, and the ability to conduct or insulate thermal energy or electric energy

Analysis of Assessed Standards

2022 – Q16

16 A student is classifying objects. Answering which question will provide the best evidence that an object is a metal?

F What is its mass?

G What is its physical state?

H Does it conduct thermal energy?

J How quickly does it dissolve in water?

***Correct Answer (H)**

Cluster	Properties of Matter
Subcluster	Physical Properties
Content	Readiness
Process	5.2(B)
Stimulus	

Data Analysis

Item	State	Local
F	12	
G	17	
H*	68	
J	3	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

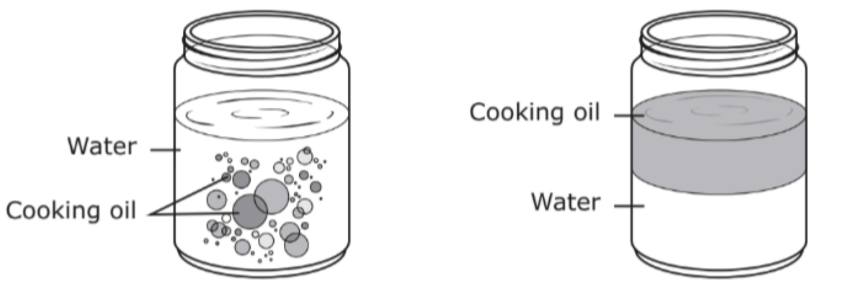
5.6(A) compare and contrast matter based on measurable, testable, or observable physical properties, including mass, magnetism, relative density (sinking and floating using water as a reference point), physical state (solid, liquid, gas), volume, solubility in water, and the ability to conduct or insulate thermal energy and electric energy

OLD **5.5(A)** classify matter based on measurable, testable, and observable physical properties, including mass, magnetism, physical state (solid, liquid, and gas), relative density (sinking and floating using water as a reference point), solubility in water, and the ability to conduct or insulate thermal energy or electric energy

Analysis of Assessed Standards

2022 – Q31

31 A student poured equal amounts of water and cooking oil into a jar. The student placed a lid on the jar and shook the mixture for five seconds and then let the jar sit for ten minutes. The results of this investigation are shown.



Immediately after shaking — Ten minutes after shaking

Which conclusion best compares a property of cooking oil and water shown in this investigation?

A Cooking oil is less dense than water.

B Water is less dense than cooking oil.

C Water dissolves in cooking oil.

D Cooking oil dissolves in water.

***Correct Answer (A)**

Cluster	Properties of Matter
Subcluster	Physical Properties
Content	Readiness
Process	5.2(D)
Stimulus	

Data Analysis

Item	State	Local
A*	78	
B	14	
C	3	
D	4	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

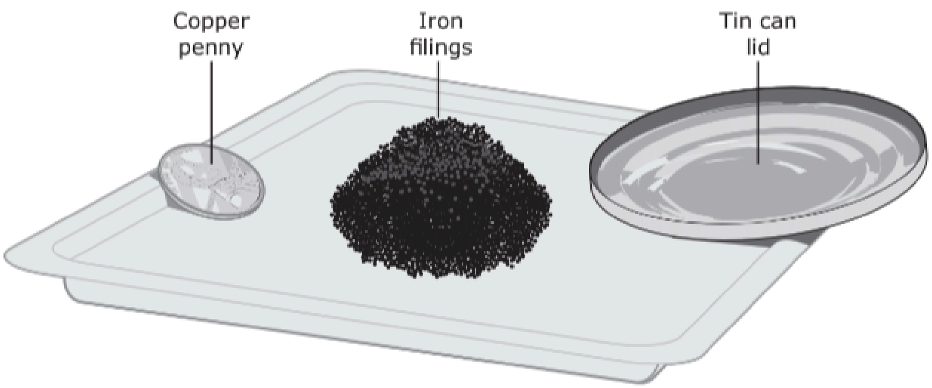
▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.6(A) compare and contrast matter based on measurable, testable, or observable physical properties, including mass, magnetism, relative density (sinking and floating using water as a reference point), physical state (solid, liquid, gas), volume, solubility in water, and the ability to conduct or insulate thermal energy and electric energy OLD **5.5(A)** classify matter based on measurable, testable, and observable physical properties, including mass, magnetism, physical state (solid, liquid, and gas), relative density (sinking and floating using water as a reference point), solubility in water, and the ability to conduct or insulate thermal energy or electric energy	**Analysis of Assessed Standards**

2022 – Q34

34 Students observe some objects on a lab tray. The students will classify the objects based on physical properties common to all of the objects.



Based on their common properties, how should these objects be classified?

F
Insulate thermal energy
Bendable
Attract to magnets

H
Conduct electrical energy
Bendable
Soluble in water

G
Soluble in water
Attract to magnets
Conduct thermal energy

J
Conduct thermal energy
Conduct electrical energy
Not soluble in water

***Correct Answer (J)**

Cluster	Properties of Matter
Subcluster	Physical Properties
Content	Readiness
Process	5.2(D)
Stimulus	

Data Analysis

Item	State	Local
F	18	
G	23	
H	7	
J*	52	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts
▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.6(A) compare and contrast matter based on measurable, testable, or observable physical properties, including mass, magnetism, relative density (sinking and floating using water as a reference point), physical state (solid, liquid, gas), volume, solubility in water, and the ability to conduct or insulate thermal energy and electric energy

OLD **5.5(A)** classify matter based on measurable, testable, and observable physical properties, including mass, magnetism, physical state (solid, liquid, and gas), relative density (sinking and floating using water as a reference point), solubility in water, and the ability to conduct or insulate thermal energy or electric energy

Analysis of Assessed Standards

2021 – Q4

4 A table of different properties of four samples of matter is shown.

Sample	Conducts Electricity	Conducts Heat	Soluble in Water	Physical State at Room Temperature
1	No	No	Yes	Solid
2	Yes	Yes	No	Solid
3	No	Yes	Yes	Liquid
4	Yes	Yes	No	Liquid

Which conclusion can be made about the samples based on the table?

F Sample 1 is made of plastic.

G Sample 2 is made of metal.

H Sample 3 is attracted to magnets.

J Sample 4 is less dense than water.

***Correct Answer (G)**

Cluster	Properties of Matter
Subcluster	Physical Properties
Content	Readiness
Process	5.2(D)
Stimulus	

Data Analysis

Item	State	Local
F	14	
G*	74	
H	4	
J	9	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

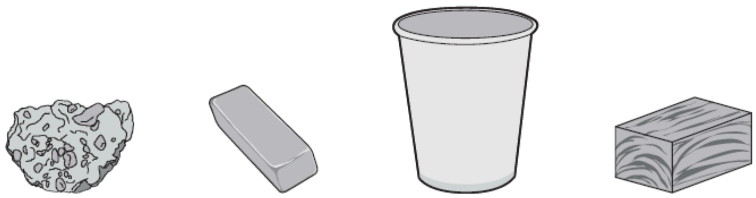
▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.6(A) compare and contrast matter based on measurable, testable, or observable physical properties, including mass, magnetism, relative density (sinking and floating using water as a reference point), physical state (solid, liquid, gas), volume, solubility in water, and the ability to conduct or insulate thermal energy and electric energy OLD **5.5(A)** classify matter based on measurable, testable, and observable physical properties, including mass, magnetism, physical state (solid, liquid, and gas), relative density (sinking and floating using water as a reference point), solubility in water, and the ability to conduct or insulate thermal energy or electric energy	**Analysis of Assessed Standards**

2021 – Q16

16 A student compares the physical properties of the four objects shown.



Which of these physical properties do all four objects have in common?

F They all have the same physical state and conduct electricity.

G They all conduct electricity and attract the same metal objects.

H They all attract the same metal objects and are not soluble in water.

J They all are not soluble in water and have the same physical state.

***Correct Answer (J)**

Cluster	Properties of Matter
Subcluster	Physical Properties
Content	Readiness
Process	
Stimulus	

Data Analysis

Item	State	Local
F	7	
G	3	
H	6	
J*	83	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.6(A) compare and contrast matter based on measurable, testable, or observable physical properties, including mass, magnetism, relative density (sinking and floating using water as a reference point), physical state (solid, liquid, gas), volume, solubility in water, and the ability to conduct or insulate thermal energy and electric energy

OLD **5.5(A)** classify matter based on measurable, testable, and observable physical properties, including mass, magnetism, physical state (solid, liquid, and gas), relative density (sinking and floating using water as a reference point), solubility in water, and the ability to conduct or insulate thermal energy or electric energy

Analysis of Assessed Standards

2021 – Q27

27 A group of students is given four small cubes of the same size and instructed to place them in a beaker of water. One cube floats to the surface of the water. Two of the cubes float in the middle of the beaker under the surface of the water. The last cube sinks to the bottom of the beaker.

Which conclusion is best supported by what the students observed?

A Two of the four cubes are soluble in water.

B All four cubes have different masses.

C Each cube is made of a different type of solid material.

D One of the cubes is more dense than the other three.

*Correct Answer (D)

Cluster	Properties of Matter
Subcluster	Physical Properties
Content	Readiness
Process	5.2(D)
Stimulus	

Data Analysis

Item	State	Local
A	8	
B	29	
C	19	
D*	44	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.6(A) compare and contrast matter based on measurable, testable, or observable physical properties, including mass, magnetism, relative density (sinking and floating using water as a reference point), physical state (solid, liquid, gas), volume, solubility in water, and the ability to conduct or insulate thermal energy and electric energy

OLD **5.5(A)** classify matter based on measurable, testable, and observable physical properties, including mass, magnetism, physical state (solid, liquid, and gas), relative density (sinking and floating using water as a reference point), solubility in water, and the ability to conduct or insulate thermal energy or electric energy

Analysis of Assessed Standards

Cluster	Properties of Matter
Subcluster	Physical Properties
Content	Readiness
Process	5.2(G)
Stimulus	

2019 – Q3

Students investigate the physical properties of some substances. They draw a table to show how the substances can be grouped. The students need to complete the table with column headings.

Physical Properties of Substances

?	?	?
• Aluminum foil • Brass key • Gold ring	• Cooking oil • Soap bubble • Wood chip • Feather	• Baking soda • Drink mix • White sugar

Which column headings should the students use for their table?

A

Good Insulators of Thermal Energy	Is Attracted by Magnets	Same Physical State

B

Good Conductors of Electrical Energy	Less Dense than Water	Soluble in Water

C

Soluble in Water	Same Physical State	Less Dense than Water

D

Is Attracted by Magnets	Good Conductors of Electrical Energy	Good Insulators of Thermal Energy

***Correct Answer (B)**

Data Analysis

Item	State	Local
A	3	
B*	89	
C	5	
D	4	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.6(A) compare and contrast matter based on measurable, testable, or observable physical properties, including mass, magnetism, relative density (sinking and floating using water as a reference point), physical state (solid, liquid, gas), volume, solubility in water, and the ability to conduct or insulate thermal energy and electric energy OLD **5.5(A)** classify matter based on measurable, testable, and observable physical properties, including mass, magnetism, physical state (solid, liquid, and gas), relative density (sinking and floating using water as a reference point), solubility in water, and the ability to conduct or insulate thermal energy or electric energy	**Analysis of Assessed Standards**

2019 – Q22

A student filled each of four beakers with 100 mL of water at 25 °C. The student added an equal amount of a different substance to each of the beakers of water.

Student Investigation

Substance	Appearance	Observations When Stirring	Observations After Stirring Stopped
Iron filings	Silvery gray	Particles swirling around	Particles settled to bottom of beaker
Papain	White powder	Cloudy changing to clear	Clear; no visible particles
Talcum powder	White powder	Floating on surface in clumps	Collected on beaker walls above liquid
Vegetable oil	Yellow liquid	Oil in clumps moving around	Formed a layer on top of water

Based on the student's observations in the table, how many of the substances did NOT dissolve in the water?

F 1 substance

G 2 substances

H 3 substances

J 4 substances

***Correct Answer (H)**

Cluster	Properties of Matter
Subcluster	Physical Properties
Content	Readiness
Process	5.2(G)
Stimulus	

Data Analysis

Item	State	Local
F	19	
G	18	
H*	54	
J	9	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.6(A) compare and contrast matter based on measurable, testable, or observable physical properties, including mass, magnetism, relative density (sinking and floating using water as a reference point), physical state (solid, liquid, gas), volume, solubility in water, and the ability to conduct or insulate thermal energy and electric energy

OLD **5.5(A)** classify matter based on measurable, testable, and observable physical properties, including mass, magnetism, physical state (solid, liquid, and gas), relative density (sinking and floating using water as a reference point), solubility in water, and the ability to conduct or insulate thermal energy or electric energy

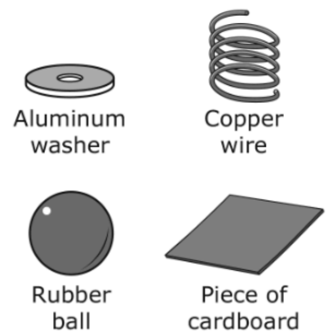
Analysis of Assessed Standards

2019 – Q30

A student wants to classify four different objects based on physical properties. The student uses the questions shown in the table to test each object.

Materials	Physical Properties		
	Insulate Thermal Energy?	Float in Water?	Conduct Electrical Energy?
1	Yes	No	No
2	No	No	Yes
3	Yes	Yes	No
4	No	No	Yes

Materials



Which statement correctly identifies two of the materials based on the classification of properties in the table?

F Material 1 is a rubber ball.
Material 2 is a piece of cardboard.

G Material 2 is an aluminum washer.
Material 3 is a copper wire.

H Material 3 is a piece of cardboard.
Material 4 is an aluminum washer.

J Material 1 is a copper wire.
Material 4 is a rubber ball.

***Correct Answer (H)**

Cluster	Properties of Matter
Subcluster	Physical Properties
Content	Readiness
Process	5.2(D)
Stimulus	

Data Analysis

Item	State	Local
F	8	
G	20	
H*	63	
J	10	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

IQ Analysis | Investigating the Question	5.6(B)	RC 1

5.6(B) demonstrate and explain that some mixtures maintain physical properties of their substances such as iron filings and sand or sand and water

2025 – Q19

A student adds 100 milliliters of water to a beaker and records the mass. The student then adds 12 grams of salt to the water and uses a hot plate to heat the mixture. The student stirs the warm water until the salt can no longer be seen in the water.

The student claims that even though the salt has disappeared, some of the properties of the salt have not changed.

What can the student do to support this claim?

Ⓐ Pour the water through a paper filter to separate out the salt

Ⓑ Add food coloring to the beaker so that the salt can be seen

Ⓒ Use a magnet to separate the salt from the water

Ⓓ Measure the mass of the water and the salt

*Correct Answer (D)
^highly chosen incorrect answer (A)

Analysis of Assessed Standards

Cluster	Properties of Matter
Subcluster	Mixtures
Content	Supporting
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A	^	
B		
C		
D*	41	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.6(B) demonstrate and explain that some mixtures maintain physical properties of their substances such as iron filings and sand or sand and water

OLD **5.5(B)** demonstrate that some mixtures maintain physical properties of their ingredients such as iron filings and sand and sand and water

2024 – Q17

A student makes a mixture of gravel, iron filings, and water. Which table correctly identifies the tools that can be used to separate each substance from the mixture?

Ⓐ

Tool	Substance Separated from Mixture
Screen	Gravel
Magnet	Iron filings
Paper filter	Water

Ⓑ

Tool	Substance Separated from Mixture
Funnel	Gravel
Paper filter	Iron filings
Hot plate	Water

Ⓒ

Tool	Substance Separated from Mixture
Screen	Gravel
Hot Plate	Iron filings
Funnel	Water

Ⓓ

Tool	Substance Separated from Mixture
Paper filter	Gravel
Magnet	Iron filings
Screen	Water

***Correct Answer (A)**
^highly chosen incorrect answer (D)

Analysis of Assessed Standards

Cluster	Properties of Matter
Subcluster	Mixtures
Content	Supporting
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A*	51	
B		
C		
D	^	

Error Analysis

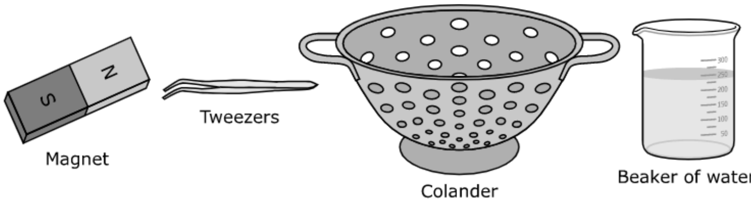
- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.6(B) demonstrate and explain that some mixtures maintain physical properties of their substances such as iron filings and sand or sand and water OLD **5.5(B)** demonstrate that some mixtures maintain physical properties of their ingredients such as iron filings and sand and sand and water	**Analysis of Assessed Standards**

2023 – Q20

The pictures show tools that can be used for separating mixtures.



Which mixture is paired with the best tool for separating the materials?

Ⓐ

Mixture	Tool
Salt and pepper	Tweezers

Ⓑ

Mixture	Tool
Pieces of cork and pebbles	Beaker of water

Ⓒ

Mixture	Tool
Safety pins and iron nails	Magnet

Ⓓ

Mixture	Tool
Sand and iron filings	Colander

***Correct Answer (B)**

Cluster	Properties of Matter
Subcluster	Mixtures
Content	Supporting
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A	9	
B*	30	
C	49	
D	13	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.6(B) demonstrate and explain that some mixtures maintain physical properties of their substances such as iron filings and sand or sand and water

OLD **5.5(B)** demonstrate that some mixtures maintain physical properties of their ingredients such as iron filings and sand and sand and water

Analysis of Assessed Standards

2022 – Q23

23 Students conduct an investigation with breakfast cereal. The first four steps of the students' investigation are in the table shown.

Breakfast Cereal Investigation

1. Grind 50 grams of cereal into a fine powder.
2. Stir the cereal powder into 500 milliliters of warm water.
3. Hold a magnet against the side of the beaker at the 250-milliliter mark.
4. Stir the mixture for three minutes.

The students are trying to determine the presence of which substance in the cereal?

A Sugar

B Iron

C Salt

D Wheat

***Correct Answer (B)**

Cluster	Properties of Matter
Subcluster	Mixtures
Content	Supporting
Process	5.4(A)
Stimulus	

Data Analysis

Item	State	Local
A	25	
B*	58	
C	4	
D	13	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.6(B) demonstrate and explain that some mixtures maintain physical properties of their substances such as iron filings and sand or sand and water

OLD **5.5(B)** demonstrate that some mixtures maintain physical properties of their ingredients such as iron filings and sand and sand and water

Analysis of Assessed Standards

2021 – Q32

32 A student mixes a sample of stones with a sample of table salt. The mass and volume of the samples were determined before mixing the samples. The mass and volume of each sample is shown.

Material	Grams (g)	Milliliters (mL)
Stones	45	25
Salt	40	35

Which statement is true about the mixture?

F The mass of the mixture is 85 grams.

G The mass of the mixture is 60 milliliters.

H The volume of the mixture is 60 grams.

J The volume of the mixture is 85 milliliters.

*Correct Answer (F)

Cluster	Properties of Matter
Subcluster	Mixtures
Content	Supporting
Process	5.2(D)
Stimulus	

Data Analysis

Item	State	Local
F*	58	
G	21	
H	12	
J	9	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.6(B) demonstrate and explain that some mixtures maintain physical properties of their substances such as iron filings and sand or sand and water

OLD **5.5(B)** demonstrate that some mixtures maintain physical properties of their ingredients such as iron filings and sand and sand and water

Analysis of Assessed Standards

Cluster	Properties of Matter
Subcluster	Mixtures
Content	Supporting
Process	
Stimulus	

2019 – Q7

The table lists the ingredients of five different mixtures.

Mixtures and Their Ingredients

Mixture	Ingredients
1	Salt, hot water, sand
2	Sugar, hot water, salt
3	Iron filings and sand
4	Pebbles, wood chips, soil
5	Powdered soap and hot water

In which mixtures do all the ingredients maintain their physical state?

A Mixtures 3 and 4 only

B Mixtures 1, 3, and 4

C Mixtures 1, 2, and 5

D Mixtures 2 and 5 only

***Correct Answer (A)**

Data Analysis

Item	State	Local
A*	77	
B	9	
C	8	
D	6	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

IQ Analysis | Investigating the Question	5.6(C)	RC 1

5.6(C) compare the properties of substances before and after they are combined into a solution and demonstrate that matter is conserved in solutions

2025 – Q12

A student added 10 grams of sugar to 500 milliliters of water while stirring the water.

Which statement **BEST** describes the sugar and water after they were mixed?

Ⓐ The sugar melted in the water, and the water stayed clear.

Ⓑ The sugar melted in the water, and the water turned white.

Ⓒ The sugar dissolved in the water, and the water stayed clear.

Ⓓ The sugar dissolved in the water, and the water turned white.

*Correct Answer (C)
^highly chosen incorrect answer (D)

Analysis of Assessed Standards	
Cluster	Properties of Matter
Subcluster	Mixtures
Content	Supporting
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A		
B		
C*	51	
D	^	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.6(C) compare the properties of substances before and after they are combined into a solution and demonstrate that matter is conserved in solutions

2025 – Q21

A student mixed lemon juice and sugar with warm water. Which property of both the lemon juice and the sugar stayed the same when they were mixed with the water?

Ⓐ Density

Ⓑ Color

Ⓒ Shape

Ⓓ Mass

*Correct Answer (D)
^highly chosen incorrect answer (A,B)

Analysis of Assessed Standards

Cluster	Properties of Matter
Subcluster	Mixtures
Content	Supporting
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A	^	
B	^	
C		
D*	43	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.6(C) compare the properties of substances before and after they are combined into a solution and demonstrate that matter is conserved in solutions

OLD **5.5(C)** identify changes that can occur in the physical properties of the ingredients of solutions such as dissolving salt in water or adding lemon juice to water

Analysis of Assessed Standards

Cluster	Properties of Matter
Subcluster	Mixtures
Content	Supporting
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

2024 – Q31

A person heats 500 milliliters of water in a beaker. Then the person stirs 10 grams of table sugar into the water.

Which property will the sugar keep after it has dissolved in the water?

Ⓐ Its grainy texture

Ⓑ Its whitish color

Ⓒ Its temperature

Ⓓ Its sweetness

Data Analysis

Item	State	Local
A		
B	^	
C		
D*	59	

Error Analysis

- ☐ Guessing
- ☐ Mixed Up Concepts
- ☐ Careless Error
- ☐ Stopped Too Early

Learning from Mistakes
Instructional Implications

***Correct Answer (D)**
^highly chosen incorrect answer (B)

5.6(C) compare the properties of substances before and after they are combined into a solution and demonstrate that matter is conserved in solutions

OLD **5.5(C)** identify changes that can occur in the physical properties of the ingredients of solutions such as dissolving salt in water or adding lemon juice to water

2023 – Q1

A student heats 250 milliliters of water in a cup to approximately 50 °C and adds 6 grams of salt. The mixture of salt and water is then stirred for 15 seconds.

Which results will the student **MOST LIKELY** observe after the mixture has been stirred for 15 seconds?

Select **TWO** correct answers.

- ☐ The salt will dissolve.
- ☐ The salt will sink to the bottom of the cup.
- ☐ The water will remain clear.
- ☐ The salt will float on the water.
- ☐ The water will change from clear to opaque white.

***Correct Answer (A, C)**

Analysis of Assessed Standards

Cluster	Properties of Matter
Subcluster	Mixtures
Content	Supporting
Process	
Item Type	Multiselect (2 pts)
Stimulus	

Data Analysis

Item	State	Local
Full Credit	**30**	
No Credit	**10**	
Partial Credit	**60**	

Error Analysis

- ☐ Guessing
- ☐ Mixed Up Concepts
- ☐ Careless Error
- ☐ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.6(C) compare the properties of substances before and after they are combined into a solution and demonstrate that matter is conserved in solutions

OLD **5.5(C)** identify changes that can occur in the physical properties of the ingredients of solutions such as dissolving salt in water or adding lemon juice to water

Analysis of Assessed Standards

Cluster	Properties of Matter
Subcluster	Mixtures
Content	Supporting
Process	
Stimulus	

2022 – Q18

18 A student pours 14 grams of sugar into a jar filled with 500 milliliters of water. The student thoroughly stirs the sugar and water to make a solution.

Which change most likely occurs to the sugar when it is added to the water?

F The sugar breaks down to form a new substance in the solution.

G The sugar changes water into a new substance in the solution.

H The sugar floats on the surface of the water in the solution.

J The sugar completely dissolves in the solution.

*Correct Answer (J)

Data Analysis

Item	State	Local
F	11	
G	8	
H	4	
J*	76	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.6(C) compare the properties of substances before and after they are combined into a solution and demonstrate that matter is conserved in solutions OLD **5.5(C)** identify changes that can occur in the physical properties of the ingredients of solutions such as dissolving salt in water or adding lemon juice to water	**Analysis of Assessed Standards**

2021 – Q21

21 A student prepared a snack that consisted of grapes, pecans, and strawberries sprinkled with white powdered sugar. The student stored the snack in a refrigerator. An hour later the student observed that the powdered sugar could no longer be seen but the fruit and nuts had not changed in appearance.

What most likely happened to the sugar in the mixture?

A The sugar evaporated at the lower temperature in the refrigerator without causing any changes to the fruit and nuts.

B The sugar was more dense than the other foods in the mixture, so it settled to the bottom of the container.

C The sugar dissolved in the moisture on the fruit.

D The sugar absorbed energy from the nuts and melted into a colorless liquid.

*Correct Answer (C)

Cluster	Properties of Matter
Subcluster	Mixtures
Content	Supporting
Process	5.2(F)
Stimulus	

Data Analysis

Item	State	Local
A	12	
B	13	
C*	70	
D	5	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.6(C) compare the properties of substances before and after they are combined into a solution and demonstrate that matter is conserved in solutions

OLD **5.5(C)** identify changes that can occur in the physical properties of the ingredients of solutions such as dissolving salt in water or adding lemon juice to water

Analysis of Assessed Standards

Cluster	Properties of Matter
Subcluster	Mixtures
Content	Supporting
Process	
Stimulus	

2019 – Q4

Which change occurs when lemon juice is mixed with water?

F The mass of the lemon juice decreases.

G The water becomes a solid.

H The lemon juice dissolves and spreads out evenly in the water.

J The volume of the water decreases.

*Correct Answer (H)

Data Analysis

Item	State	Local
F	6	
G	1	
H*	87	
J	6	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

IQ Analysis | Investigating the Question	3.6(C)	RC 1

3.6(C) predict, observe, and record changes in the state of matter caused by heating or cooling in a variety of substances such as ice becoming liquid water, condensation forming on the outside of a glass, or liquid water being heated to the point of becoming water vapor (gas)

2025 – Q8

Students conduct an investigation to determine how water changes state when it is heated and cooled. They complete the steps shown in the table.

Step	Temperature Condition	Observation
1	Place the beaker of ice cubes outside on a 25 °C, sunny day for 2 hours.	The ice cubes disappear, and only liquid water is present in the beaker.
2	Bring the beaker of water into an air-conditioned room for 10 minutes.	Water droplets form on the outside of the beaker.
3	Heat the beaker of water on a hot plate at 100 °C for 10 minutes.	?

Which observation will the students **MOST LIKELY** make in Step 3?

Ⓐ Water vapor forms from outside of the beaker, and the amount of water in the beaker decreases.

Ⓑ The water in the beaker begins to form a layer of ice on top.

Ⓒ The water in the beaker begins bubbling, and the amount of water decreases.

Ⓓ Water droplets begin to form on the inside of the beaker.

***Correct Answer (C)**
^highly chosen incorrect answer (A)

Analysis of Assessed Standards

Cluster	Properties of Matter
Subcluster	Physical Properties
Content	Supporting
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A	^	
B		
C*	43	
D		

Error Analysis

- ☐ Guessing
- ☐ Mixed Up Concepts
- ☐ Careless Error
- ☐ Stopped Too Early

Learning from Mistakes
Instructional Implications

3.6(C) predict, observe, and record changes in the state of matter caused by heating or cooling in a variety of substances such as ice becoming liquid water, condensation forming on the outside of a glass, or liquid water being heated to the point of becoming water vapor (gas)

OLD **3.5(C)** predict, observe, and record changes in the state of matter caused by heating or cooling such as ice becoming liquid water or condensation forming on the outside of a glass of ice water or liquid water being heated to the point of becoming water vapor

Analysis of Assessed Standards

2023 – Q6

Which statement **BEST** describes what happens as water changes from one state to another?

Ⓐ As water is cooled, it freezes and changes from a solid to a liquid.

Ⓑ As water is cooled, it condenses and changes from a gas to a liquid.

Ⓒ As water is heated, it evaporates and changes from a liquid to a solid.

Ⓓ As water is heated, it melts and changes from a gas to a solid.

*Correct Answer (B)

Cluster	Properties of Matter
Subcluster	Physical Properties
Content	Supporting
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A	26	
B*	57	
C	12	
D	5	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

3.6(C) predict, observe, and record changes in the state of matter caused by heating or cooling in a variety of substances such as ice becoming liquid water, condensation forming on the outside of a glass, or liquid water being heated to the point of becoming water vapor (gas)

OLD **3.5(C)** predict, observe, and record changes in the state of matter caused by heating or cooling such as ice becoming liquid water or condensation forming on the outside of a glass of ice water or liquid water being heated to the point of becoming water vapor

Analysis of Assessed Standards

Cluster	Properties of Matter
Subcluster	Physical Properties
Content	Supporting
Process	
Stimulus	

2021 – Q11

11 Students observe a glass of ice and water. The glass is dry on the outside. After ten minutes students see drops of water on the outside of the glass.

Which statement best explains the students' observations?

A The water vapor in the air changes to a liquid when it touches the cold glass.

B The cold water inside the glass rises and then slides down the outside of the glass.

C The ice in the glass melted and caused the water to overflow the glass.

D The water moved to the outside surface through tiny holes in the glass.

Data Analysis

Item	State	Local
A*	70	
B	12	
C	14	
D	4	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

***Correct Answer (A)**

3.6(C) predict, observe, and record changes in the state of matter caused by heating or cooling in a variety of substances such as ice becoming liquid water, condensation forming on the outside of a glass, or liquid water being heated to the point of becoming water vapor (gas)

OLD **3.5(C)** predict, observe, and record changes in the state of matter caused by heating or cooling such as ice becoming liquid water or condensation forming on the outside of a glass of ice water or liquid water being heated to the point of becoming water vapor

Analysis of Assessed Standards

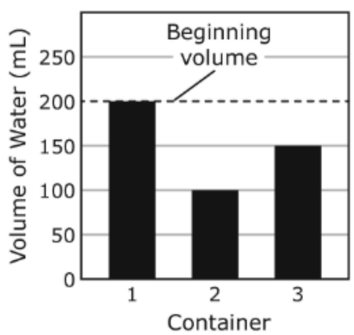
2019 – Q15

Students plan to investigate the different states of matter. They will measure the volume of water in three containers after five days. They make this table to organize the data.

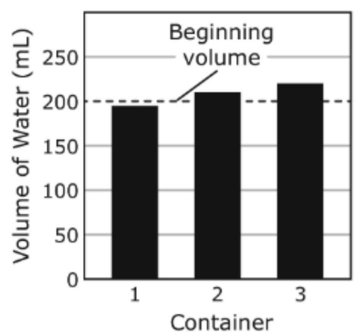
Container	Volume of Water (beginning)	Temperature of Water (beginning)	Location of Container	Volume of Water (after five days)
1	200 mL	24°C	Inside a refrigerator	?
2	200 mL	24°C	On a desk in front of low-speed fan	?
3	200 mL	24°C	On a sunny windowsill in front of an opened window	?

Which graph represents the volume of water most likely left in the containers after five days?

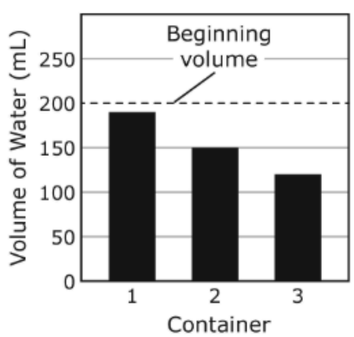
A



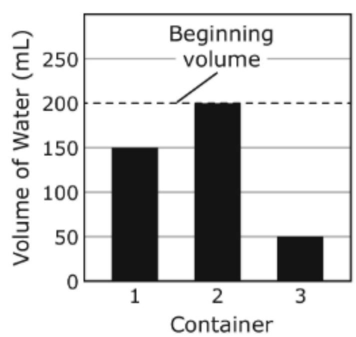
C



B



D



***Correct Answer (B)**

Cluster	Properties of Matter
Subcluster	Physical Properties
Content	Supporting
Process	5.2(G)
Stimulus	

Data Analysis

Item	State	Local
A	15	
B*	54	
C	11	
D	20	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

Force and Motion

5.7 Force, motion, and energy. The student knows the nature of forces and the patterns of their interactions.

IQ Analysis | Investigating the Question	5.7(B)	RC 2

5.7(B) design a simple experimental investigation that tests the effect of force on an object in a system such as a car on a ramp or a balloon rocket on a string

2025 – Q28

Students design an investigation to test the strength of different magnets. They use three magnets and steel paper clips. Each magnet is placed at a different distance from the steel paper clips. The table they will use to record their data is shown.

Magnet	Distance from Steel Paper Clips (centimeters)	Number of Steel Paper Clips Attracted
1	15	
2	30	
3	45	

What should the students do to improve the investigation?

Ⓐ Use different steel objects for each magnet

Ⓑ Place the paper clips farther away from each magnet

Ⓒ Place each magnet at the same distance from the paper clips

Ⓓ Repeat the investigation using plastic paper clips

***Correct Answer (C)**
^highly chosen incorrect answer (A)

Analysis of Assessed Standards

Cluster	Force and Motion
Subcluster	Investigating Forces
Content	Supporting
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A	^	
B		
C*	40	
D		

Error Analysis

☐ Guessing ☐ Mixed Up Concepts

☐ Careless Error ☐ Stopped Too Early

Learning from Mistakes Instructional Implications

5.7(B) design a simple experimental investigation that tests the effect of force on an object in a system such as a car on a ramp or a balloon rocket on a string OLD **5.6(D)** design a simple experimental investigation that tests the effect of force on an object	**Analysis of Assessed Standards**

2024 – Q19

A student wants to investigate how the force with which a ball is thrown affects how high of a splash it makes in a pool.

Which table **BEST** represents how the investigation should be set up?

Ⓐ

Independent variable	Type of ball
Dependent variable	Weight of the ball
Controlled variables	Height of the splash Force the ball is thrown with Height the ball is thrown from

Ⓑ

Independent variable	Height of the splash
Dependent variable	Force the ball is thrown with
Controlled variables	Weight of the ball Type of ball Height the ball is thrown from

Ⓒ

Independent variable	Weight of the ball
Dependent variable	Type of ball
Controlled variables	Height of the splash Force the ball is thrown with Height the ball is thrown from

Ⓓ

Independent variable	Force the ball is thrown with
Dependent variable	Height of the splash
Controlled variables	Weight of the ball Type of ball Height the ball is thrown from

***Correct Answer (D)**
^highly chosen incorrect answer (A)

Cluster	Force and Motion
Subcluster	Investigating Forces
Content	Supporting
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A	^	
B		
C		
D*	22	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

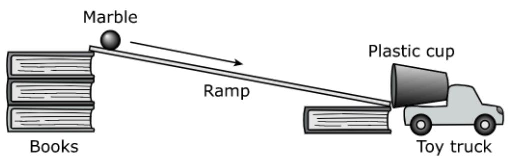
5.7(B) design a simple experimental investigation that tests the effect of force on an object in a system such as a car on a ramp or a balloon rocket on a string

OLD **5.6(D)** design a simple experimental investigation that tests the effect of force on an object

2023 – Q15

A class is using the setup shown to test this hypothesis.

> The shorter the stack of books becomes, the shorter the distance the toy truck will roll.



What is **ONE** more piece of equipment needed to test the hypothesis **AND** how can the hypothesis be tested?

Look at the diagram carefully. Then enter your answer and explanation in the box provided.

Chars 0/475

***Correct Answer (The student needs to identify at least one more piece of equipment such as a ruler, measuring tape, or meter stick AND include at least one idea for how the hypothesis can be tested: varying the size of the book stack, measuring distance traveled, repeated trials.)**

Analysis of Assessed Standards

Cluster	Force and Motion
Subcluster	Investigating Forces
Content	Supporting
Process	
Item Type	Short Constructed Response (2 pts)
Stimulus	

Data Analysis

Item	State	Local
Full Credit	**27**	
No Credit	**55**	
Partial Credit	**18**	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.7(B) design a simple experimental investigation that tests the effect of force on an object in a system such as a car on a ramp or a balloon rocket on a string

OLD **5.6(D)** design a simple experimental investigation that tests the effect of force on an object

2022 – Q36

36 Students conduct experiments to investigate friction. Which experiment will best test the effect of friction on objects?

F Drop two balls from the same height at the same time

G Roll two marbles on the carpet from the same starting point at the same time and with the same amount of force

H Roll three marbles across three different surfaces from the same starting point at the same time and with the same amount of force

J Release two balls from the top of a ramp at the same time

***Correct Answer (H)**

Analysis of Assessed Standards

Cluster	Force and Motion
Subcluster	Investigating Forces
Content	Supporting
Process	5.3(A)
Stimulus	

Data Analysis

Item	State	Local
F	8	
G	20	
H*	63	
J	10	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

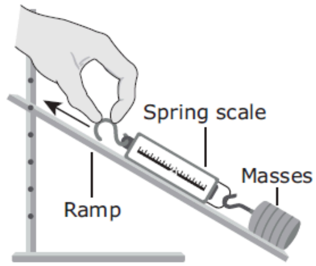
Learning from Mistakes
Instructional Implications

5.7(B) design a simple experimental investigation that tests the effect of force on an object in a system such as a car on a ramp or a balloon rocket on a string

OLD **5.6(D)** design a simple experimental investigation that tests the effect of force on an object

2021 – Q29

29 Students investigate force. The masses they use begin at rest on the ramp. The setup the students use is shown.



Which change will reduce the amount of force needed to move the masses?

A Decrease the height of the ramp

B Increase the height of the ramp

C Add an additional mass

D Pull the spring scale with two hands

***Correct Answer (A)**

Analysis of Assessed Standards

Cluster	Force and Motion
Subcluster	Investigating Forces
Content	Supporting
Process	5.2(A)
Stimulus	

Data Analysis

Item	State	Local
A*	46	
B	22	
C	13	
D	19	

Error Analysis

- ☐ Guessing
- ☐ Mixed Up Concepts
- ☐ Careless Error
- ☐ Stopped Too Early

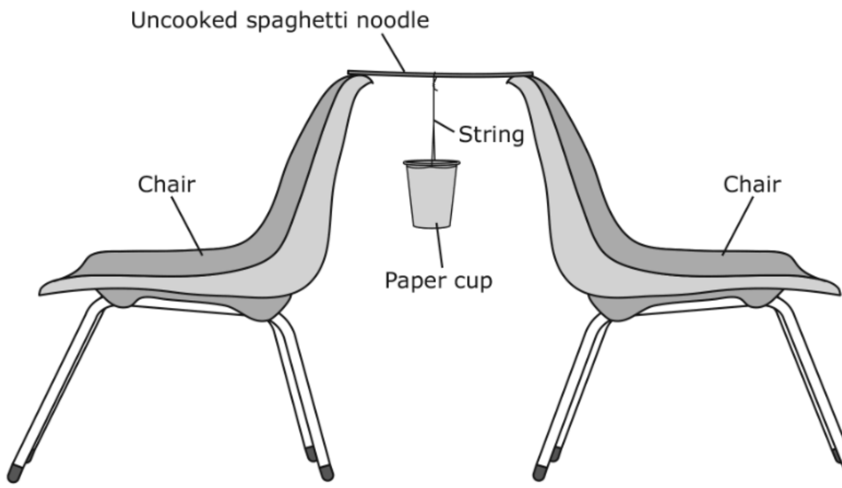
Learning from Mistakes
Instructional Implications

5.7(B) design a simple experimental investigation that tests the effect of force on an object in a system such as a car on a ramp or a balloon rocket on a string

OLD **5.6(D)** design a simple experimental investigation that tests the effect of force on an object

2019 – Q11

A student conducts the investigation shown in the diagram. In this experiment a paper cup hangs from a string tied to a single uncooked spaghetti noodle. The student measures and records the mass of a penny. The student then adds pennies to the paper cup one at a time.



Which question is the student most likely trying to answer with this investigation?

A How many spaghetti noodles will it take to hold up the mass of a penny?

B How much force will it take to break the spaghetti noodle?

C How long should the string that holds the paper cup be in order to support the greatest mass of pennies?

D How does the distance between the two chairs affect the amount of force it takes for the spaghetti noodle to break?

***Correct Answer (B)**

Analysis of Assessed Standards

Cluster	Force and Motion
Subcluster	Investigating Forces
Content	Supporting
Process	5.2(B)
Stimulus	

Data Analysis

Item	State	Local
A	7	
B*	74	
C	12	
D	7	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

IQ Analysis | Investigating the Question	3.7(B)	RC 2

3.7(B) plan and conduct a descriptive investigation to demonstrate and explain how position and motion can be changed by pushing and pulling objects such as swings, balls, and wagons

2025 – Q31

A student is helping a teacher move a bookcase. The student pushes on the bookcase, but the bookcase does not move.

Which action should the student take in order to move the bookcase?

Ⓐ Apply more force to one side of the bookcase

Ⓑ Apply more downward force to the bookcase

Ⓒ Apply less force to the bookcase

Ⓓ Apply equal force to all sides of the bookcase

***Correct Answer (A)**
^highly chosen incorrect answer (D)

Analysis of Assessed Standards

Cluster	Force and Motion
Subcluster	Investigating Forces
Content	Supporting
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A*	67	
B		
C		
D	^	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

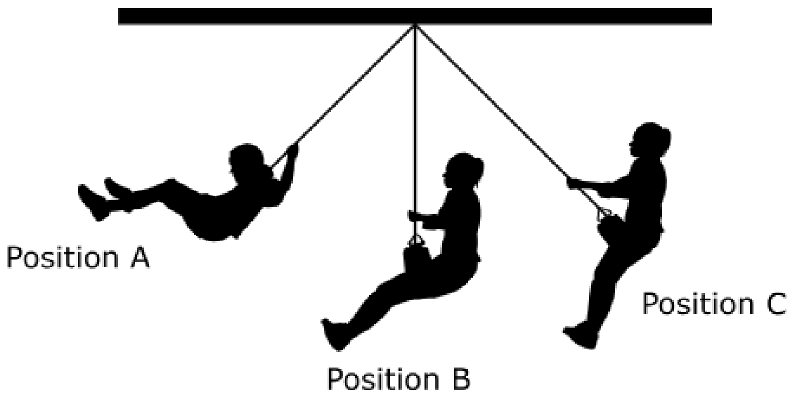
Learning from Mistakes
Instructional Implications

3.7(B) plan and conduct a descriptive investigation to demonstrate and explain how position and motion can be changed by pushing and pulling objects such as swings, balls, and wagons

OLD **3.6(B)** demonstrate and observe how position and motion can be changed by pushing and pulling objects such as swings, balls, and wagons

2024 – Q26

A diagram of a student on a swing is shown.



Which statement correctly describes the forces that affect the student's position and motion?

Ⓐ The force of gravity causes the student to speed up from Position A to Position B.

Ⓑ The force of gravity causes the student to speed up from Position B to Position C.

Ⓒ The force of the student's legs moving causes the motion to stop at Position B.

Ⓓ The force of the student pulling on the swing causes the motion to stop at Position A.

***Correct Answer (A)**
^highly chosen incorrect answer (B)

Analysis of Assessed Standards

Cluster	Force and Motion
Subcluster	Investigating Forces
Content	Supporting
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A*	39	
B	^	
C		
D		

Error Analysis
▢ Guessing ▢ Mixed Up Concepts
▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

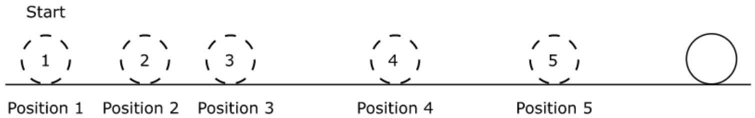
3.7(B) plan and conduct a descriptive investigation to demonstrate and explain how position and motion can be changed by pushing and pulling objects such as swings, balls, and wagons

OLD **3.6(B)** demonstrate and observe how position and motion can be changed by pushing and pulling objects such as swings, balls, and wagons

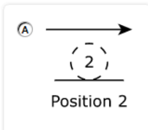
2023 – Q14

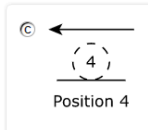
A student pushes a ball along a hallway. The diagram shows the position of the ball every second for five seconds after it starts rolling. At one of the positions shown, an extra force is applied to the ball as it moves.

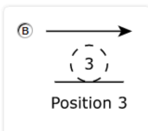
Different Positions of a Rolling Ball

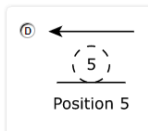


Which diagram **BEST** represents the direction of the extra force and the position where the extra force was applied to the ball?









***Correct Answer (B)**

Analysis of Assessed Standards

Cluster	Force and Motion
Subcluster	Investigating Forces
Content	Supporting
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A	15	
B*	60	
C	13	
D	12	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

3.7(B) plan and conduct a descriptive investigation to demonstrate and explain how position and motion can be changed by pushing and pulling objects such as swings, balls, and wagons

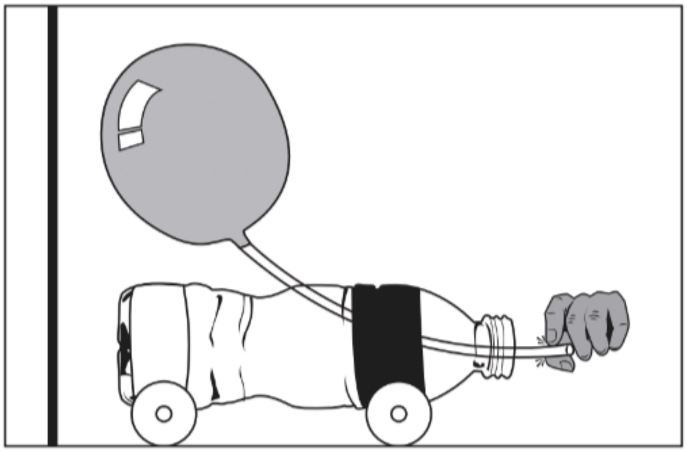
OLD **3.6(B)** demonstrate and observe how position and motion can be changed by pushing and pulling objects such as swings, balls, and wagons

Analysis of Assessed Standards

Cluster	Force and Motion
Subcluster	Investigating Forces
Content	Supporting
Process	5.2(D)
Stimulus	

2022 – Q33

33 Students build a balloon-rocket car using a balloon, a straw, a water bottle, and some tape. When the straw is released by a student, the air from the balloon comes out. A diagram of the car is shown.



Which statement best describes the motion of the car when the straw is released?

A The car will move away from the wall in the same direction as the air that is leaving the balloon.

B The car will move toward the wall because of the force of the air from the straw.

C The car will move fast because the straw gets light as the air is released.

D The car will not move because the air escapes from the balloon.

*Correct Answer (B)

Data Analysis

Item	State	Local
A	23	
B*	59	
C	9	
D	8	

Error Analysis

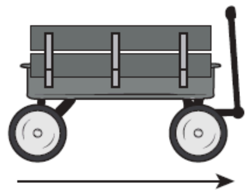
- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

3.7(B) plan and conduct a descriptive investigation to demonstrate and explain how position and motion can be changed by pushing and pulling objects such as swings, balls, and wagons OLD **3.6(B)** demonstrate and observe how position and motion can be changed by pushing and pulling objects such as swings, balls, and wagons	**Analysis of Assessed Standards**

2021 – Q5

5 A wagon is pushed and begins to move. As the wagon moves, it slows and comes to a stop. The wagon and the direction it is pushed are shown.



What force causes the wagon to stop?

A The force of gravity which is acting in the same direction as the arrow

B The force of friction which is acting in the same direction as the arrow

C The force of gravity which is acting in the opposite direction of the arrow

D The force of friction which is acting in the opposite direction of the arrow

***Correct Answer (D)**

Cluster	Force and Motion
Subcluster	Investigating Forces
Content	Supporting
Process	5.2(D)
Stimulus	

Data Analysis

Item	State	Local
A	10	
B	21	
C	16	
D*	53	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

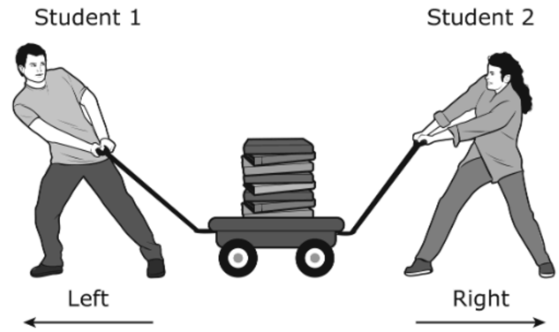
▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

3.7(B) plan and conduct a descriptive investigation to demonstrate and explain how position and motion can be changed by pushing and pulling objects such as swings, balls, and wagons OLD **3.6(B)** demonstrate and observe how position and motion can be changed by pushing and pulling objects such as swings, balls, and wagons	**Analysis of Assessed Standards**

2019 – Q6

Students fill a cart with books. The cart has a handle on each end.



Which actions will make moving the cart in one direction easiest for the students?

F Student 1 pulls the cart to the left while Student 2 pulls the cart to the right.

G Student 1 pulls the cart to the left while Student 2 pushes the cart to the left.

H Student 1 pushes the cart to the right while Student 2 pushes the cart downward.

J Student 1 pushes the cart to the right while Student 2 pulls the cart upward.

***Correct Answer (G)**

Cluster	Force and Motion
Subcluster	Investigating Forces
Content	Supporting
Process	
Stimulus	

Data Analysis

Item	State	Local
F	10	
G*	83	
H	2	
J	4	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

Energy

5.8 Force, motion, and energy. The student knows that energy is everywhere and can be observed in cycles, patterns, and systems.

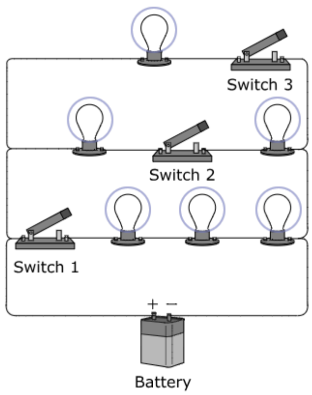
IQ Analysis | Investigating the Question	5.8(B)	RC 2

5.8(B) demonstrate that electrical energy in complete circuits can be transformed into motion, light, sound, or thermal energy and identify the requirements for a functioning electrical circuit

2025 – Q11

In the electric circuit shown, which lightbulbs will shine when Switch 1 is open and Switches 2 and 3 are closed?

Select **THREE** correct answers.



***Correct Answer (the three bulbs on the top two lines)**

Analysis of Assessed Standards

Cluster	Energy
Subcluster	Electrical Energy
Content	Readiness
Process	
Item Type	Hot Spot (2 pts)
Stimulus	

Data Analysis

Item	State	Local
Full Credit	70	
No Credit	21	
Partial Credit	9	

Error Analysis

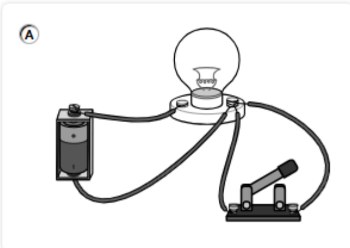
- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

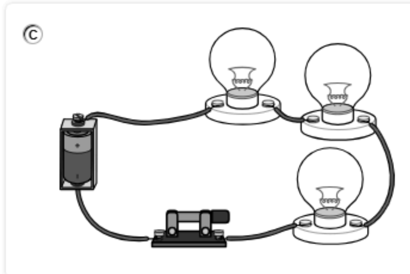
Learning from Mistakes Instructional Implications

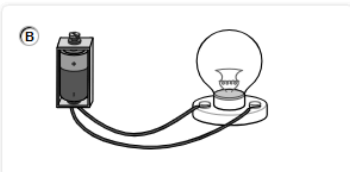
5.8(B) demonstrate that electrical energy in complete circuits can be transformed into motion, light, sound, or thermal energy and identify the requirements for a functioning electrical circuit

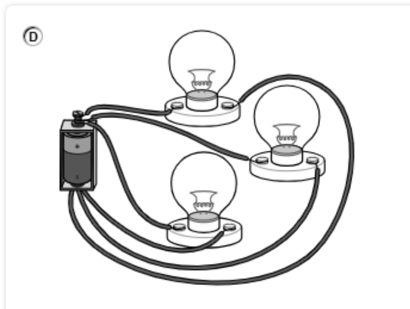
2025 – Q18

Which circuit is **NOT** a complete circuit?









*Correct Answer (B)
^highly chosen incorrect answer (A)

Analysis of Assessed Standards

Cluster	Energy
Subcluster	Electrical Energy
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A	^	
B*	52	
C		
D		

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

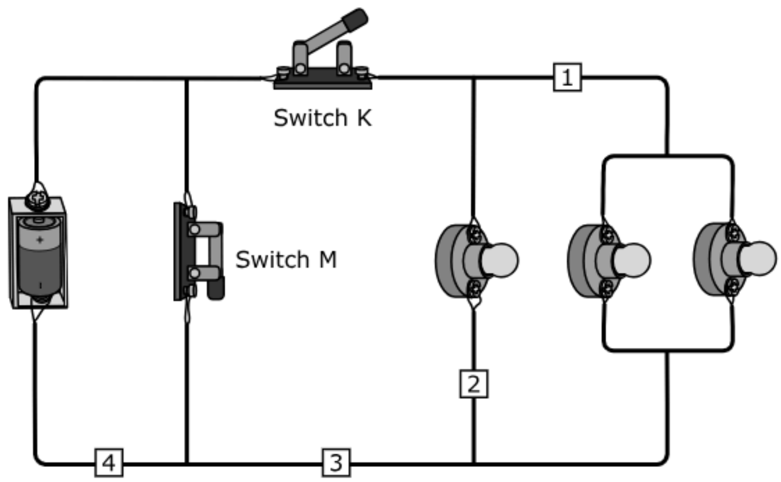
▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.8(B) demonstrate that electrical energy in complete circuits can be transformed into motion, light, sound, or thermal energy and identify the requirements for a functioning electrical circuit

2025 – Q22

A group of students build the circuit shown.



The students want to add an electric motor that still functions when Switch K is open.

At which location should the electric motor be placed?

Ⓐ Location 1

Ⓑ Location 2

Ⓒ Location 3

Ⓓ Location 4

*Correct Answer (D)
^highly chosen incorrect answer (C)

Analysis of Assessed Standards

Cluster	Energy
Subcluster	Electrical Energy
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A		
B		
C	^	
D*	45	

Error Analysis
- ☐ Guessing
- ☐ Mixed Up Concepts
- ☐ Careless Error
- ☐ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.8(B) demonstrate that electrical energy in complete circuits can be transformed into motion, light, sound, or thermal energy and identify the requirements for a functioning electrical circuit

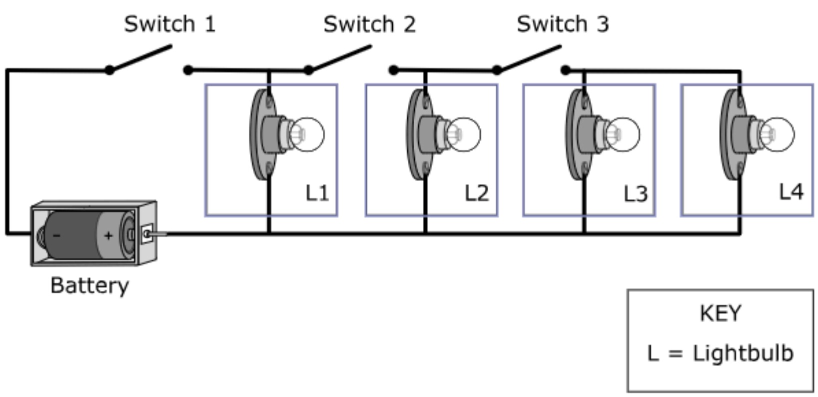
OLD **5.6(B)** demonstrate that the flow of electricity in closed circuits can produce light, heat, or sound

2024 – Q5

A student builds the circuit shown. The student closes Switch 1 and Switch 3.

Which lightbulb will glow?

Select **ONE** correct answer.



*Correct Answer (L1)

Analysis of Assessed Standards

Cluster	Energy
Subcluster	Electrical Energy
Content	Readiness
Process	
Item Type	Hot Spot (1 pt)
Stimulus	

Data Analysis

Item	State	Local
Full Credit	56	
No Credit	44	

Error Analysis

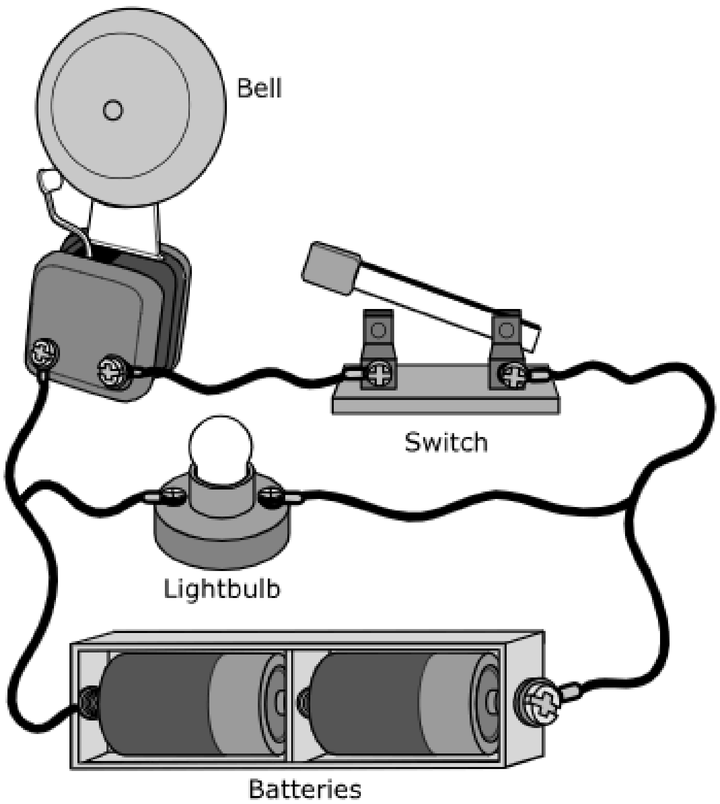
- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.8(B) demonstrate that electrical energy in complete circuits can be transformed into motion, light, sound, or thermal energy and identify the requirements for a functioning electrical circuit OLD **5.6(B)** demonstrate that the flow of electricity in closed circuits can produce light, heat, or sound	**Analysis of Assessed Standards**

2024 – Q18

Students create the circuit shown.



What do the lightbulb and the bell do when the switch is left open?

Ⓐ The lightbulb glows, and the bell rings.

Ⓑ The lightbulb glows, and the bell does not ring.

Ⓒ The lightbulb does not glow, and the bell rings.

Ⓓ The lightbulb does not glow, and the bell does not ring.

***Correct Answer (B)**
^highly chosen incorrect answer (D)

Cluster	Energy
Subcluster	Electrical Energy
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A		
B*	51	
C		
D	^	

Error Analysis
- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

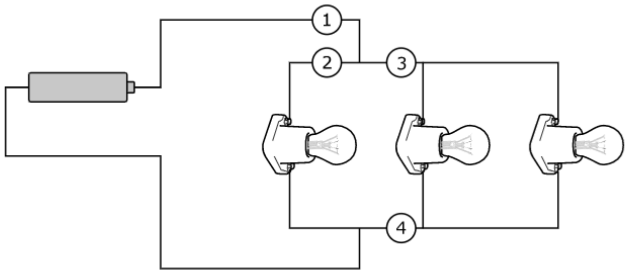
Learning from Mistakes
Instructional Implications

5.8(B) demonstrate that electrical energy in complete circuits can be transformed into motion, light, sound, or thermal energy and identify the requirements for a functioning electrical circuit

OLD **5.6(B)** demonstrate that the flow of electricity in closed circuits can produce light, heat, or sound

2023 – Q5

Students made the circuit shown.



Cutting the wire at which numbered position would allow two of the lightbulbs to remain lit?

Ⓐ Position 1

Ⓑ Position 2

Ⓒ Position 3

Ⓓ Position 4

***Correct Answer (B)**

Analysis of Assessed Standards

Cluster	Energy
Subcluster	Electrical Energy
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A	10	
B*	59	
C	13	
D	17	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

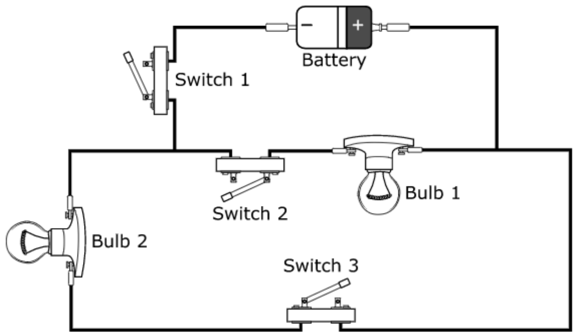
Learning from Mistakes
Instructional Implications

5.8(B) demonstrate that electrical energy in complete circuits can be transformed into motion, light, sound, or thermal energy and identify the requirements for a functioning electrical circuit

OLD **5.6(B)** demonstrate that the flow of electricity in closed circuits can produce light, heat, or sound

2023 – Q29

A group of students builds the electric circuit shown.



Which switches must be closed for Bulb 1 to remain dark but Bulb 2 to produce light?

Ⓐ Switch 1 only must be closed.

Ⓑ Switches 2 and 3 must be closed.

Ⓒ Switches 1 and 3 must be closed.

Ⓓ Switches 1, 2, and 3 must be closed.

***Correct Answer (C)**

Analysis of Assessed Standards

Cluster	Energy
Subcluster	Electrical Energy
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A	**16**	
B	**15**	
C*	**60**	
D	**9**	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

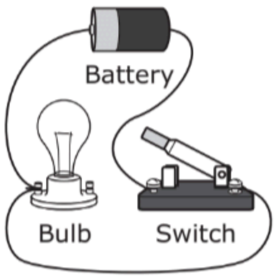
5.8(B) demonstrate that electrical energy in complete circuits can be transformed into motion, light, sound, or thermal energy and identify the requirements for a functioning electrical circuit

OLD **5.6(B)** demonstrate that the flow of electricity in closed circuits can produce light, heat, or sound

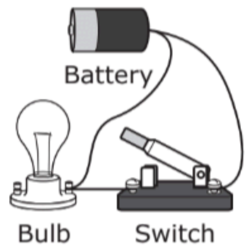
2022 – Q1

1 Which circuit shown will produce light when the switch is closed?

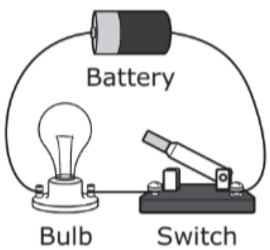
A



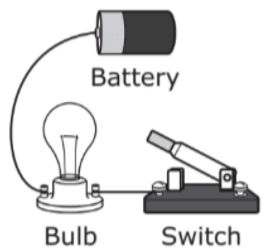
C



B



D



***Correct Answer (B)**

Analysis of Assessed Standards

Cluster	Energy
Subcluster	Electrical Energy
Content	Readiness
Process	5.2(D)
Stimulus	

Data Analysis

Item	State	Local
A	7	
B*	87	
C	3	
D	3	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.8(B) demonstrate that electrical energy in complete circuits can be transformed into motion, light, sound, or thermal energy and identify the requirements for a functioning electrical circuit

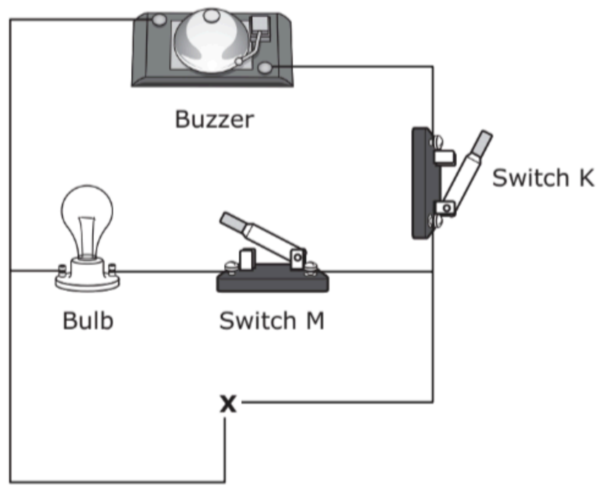
OLD **5.6(B)** demonstrate that the flow of electricity in closed circuits can produce light, heat, or sound

Analysis of Assessed Standards

Cluster	Energy
Subcluster	Electrical Energy
Content	Readiness
Process	5.2(B)
Stimulus	

2022 – Q24

24 A group of students is building the circuit represented in the diagram.



What object can be used at Position X for the buzzer to sound when Switch K is closed?

F A power source

G An insulated wire

H Another bulb

J Another switch

***Correct Answer (F)**

Data Analysis

Item	State	Local
F*	71	
G	11	
H	11	
J	8	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

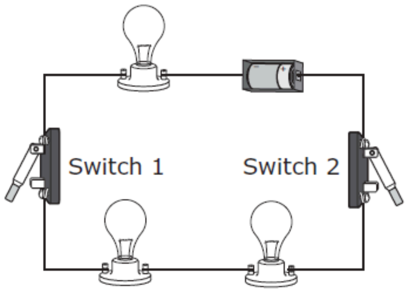
Learning from Mistakes Instructional Implications

5.8(B) demonstrate that electrical energy in complete circuits can be transformed into motion, light, sound, or thermal energy and identify the requirements for a functioning electrical circuit

OLD **5.6(B)** demonstrate that the flow of electricity in closed circuits can produce light, heat, or sound

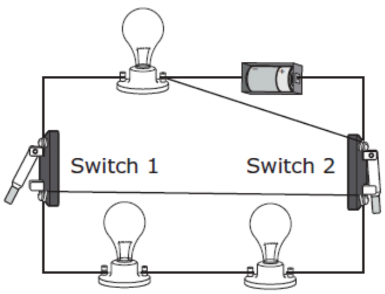
2021 – Q8

8 A student wants to make a change to the circuit shown below so that when Switch 1 is open and Switch 2 is closed, only one light will be on in the circuit.

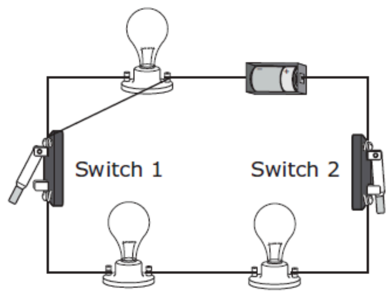


Which diagram shows how the student should change the circuit?

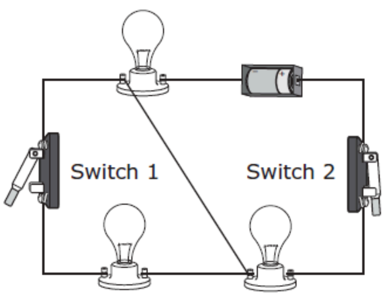
F



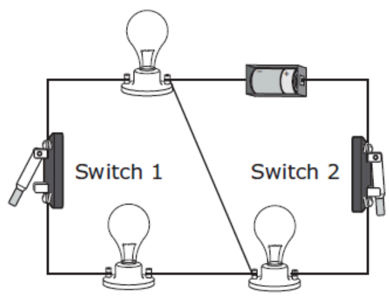
H



G



J



***Correct Answer (J)**

Analysis of Assessed Standards

Cluster	Energy
Subcluster	Electrical Energy
Content	Readiness
Process	5.2(D)
Stimulus	

Data Analysis

Item	State	Local
F	31	
G	16	
H	18	
J*	35	

Error Analysis

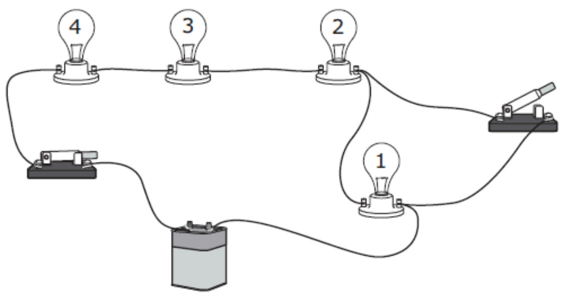
- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.8(B) demonstrate that electrical energy in complete circuits can be transformed into motion, light, sound, or thermal energy and identify the requirements for a functioning electrical circuit OLD **5.6(B)** demonstrate that the flow of electricity in closed circuits can produce light, heat, or sound	**Analysis of Assessed Standards**

2021 – Q20

20 A student constructs the circuit shown for a science demonstration.



With the switches in these positions, which lights are on?

F All the lights

G Lights 1 and 2 only

H Lights 3 and 4 only

J None of the lights

***Correct Answer (F)**

Cluster	Energy
Subcluster	Electrical Energy
Content	Readiness
Process	5.2(D)
Stimulus	

Data Analysis

Item	State	Local
F*	57	
G	15	
H	20	
J	8	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.8(B) demonstrate that electrical energy in complete circuits can be transformed into motion, light, sound, or thermal energy and identify the requirements for a functioning electrical circuit

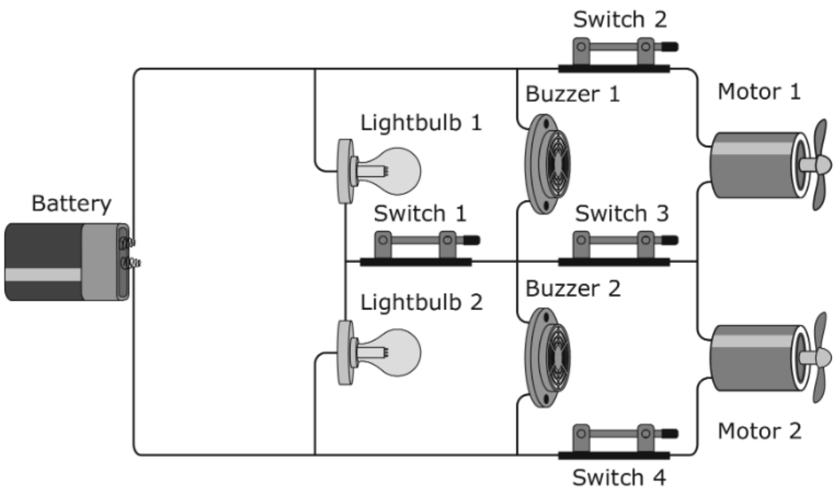
OLD **5.6(B)** demonstrate that the flow of electricity in closed circuits can produce light, heat, or sound

Analysis of Assessed Standards

Cluster	Energy
Subcluster	Electrical Energy
Content	Readiness
Process	5.2(D)
Stimulus	

2019 – Q12

A student builds a circuit allowing the lightbulbs to light, the buzzers to make sound, and the motors to turn.



Which two switches can be open and still allow all of the parts to work?

F Switches 1 and 3

G Switches 1 and 4

H Switches 2 and 3

J Switches 3 and 4

***Correct Answer (F)**

Data Analysis

Item	State	Local
F*	74	
G	8	
H	9	
J	8	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

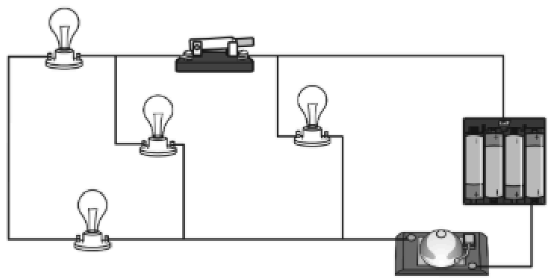
5.8(B) demonstrate that electrical energy in complete circuits can be transformed into motion, light, sound, or thermal energy and identify the requirements for a functioning electrical circuit OLD **5.6(B)** demonstrate that the flow of electricity in closed circuits can produce light, heat, or sound	**Analysis of Assessed Standards**

2019 – Q16

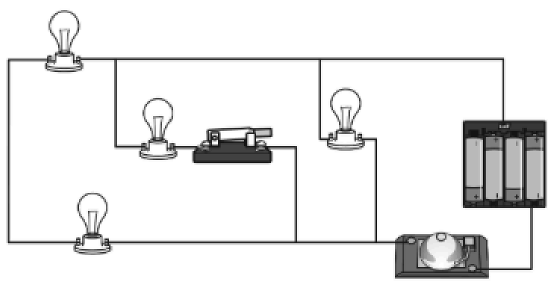
A student wants to build a circuit with four lightbulbs and one bell. The student wants to place a switch in the circuit so that only one light will still be on and the bell will still ring when the switch is opened.

Which of these circuits should the student build?

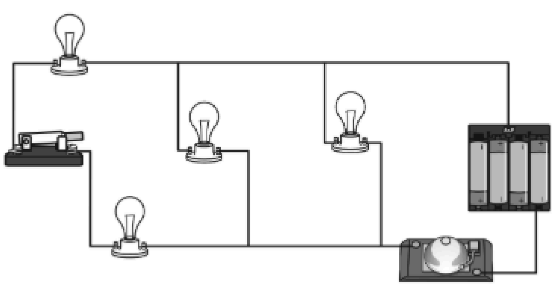
F



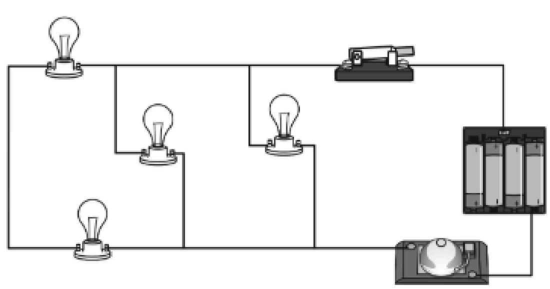
G



H



J



***Correct Answer (F)**

Cluster	Energy
Subcluster	Electrical Energy
Content	Readiness
Process	5.2(D)
Stimulus	

Data Analysis

Item	State	Local
F*	42	
G	26	
H	15	
J	17	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

IQ Analysis | Investigating the Question	5.8(C)	RC 2

5.8(C) demonstrate and explain how light travels in a straight line and can be reflected, refracted, or absorbed

2025 – Q2

Students place a toy elephant on the floor in a room without windows. The students turn the room's light on and off and record their observations in the table.

Light	Observation
On	Can see the toy
Off	Cannot see the toy

Which statement describes the conditions that allow the students to see the toy?

Ⓐ Light rays reflect off the toy and travel to the lightbulb.

Ⓑ Light rays reflect off the toy and travel to the students' eyes.

Ⓒ Light rays refract off the toy and travel to the lightbulb.

Ⓓ Light rays refract off the toy and travel to the students' eyes.

*Correct Answer (B)
^highly chosen incorrect answer (D)

Analysis of Assessed Standards

Cluster	Energy
Subcluster	Light
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A		
B*	60	
C		
D	^	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

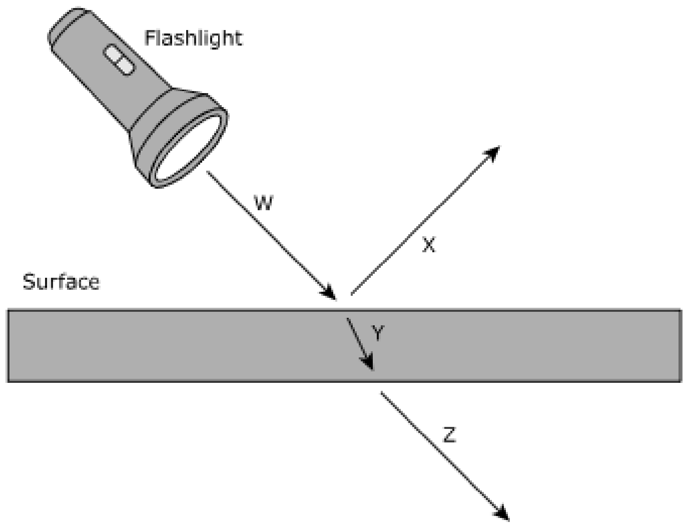
Learning from Mistakes
Instructional Implications

5.8(C) demonstrate and explain how light travels in a straight line and can be reflected, refracted, or absorbed

2025 – Q17

This question has two parts. First, answer Part A. Then answer Part B.

The diagram shows how a ray of light behaves when it strikes a surface.



Part A
Which statement describes a behavior of light shown in the diagram?

Ⓐ Arrow W and Arrow X show light refracting as it interacts with the surface.

Ⓑ Arrow W and Arrow Y show light refracting as it interacts with the surface.

Ⓒ Arrow Y and Arrow Z show light reflecting as it interacts with the surface.

Ⓓ Arrow W and Arrow Z show light reflecting as it interacts with the surface.

Part B
Which description of the surface **BEST** supports the answer to Part A?

Ⓐ The surface is a shiny piece of metal.

Ⓑ The surface is a dull piece of metal.

Ⓒ The surface is a clear piece of glass.

Ⓓ The surface is a hard piece of wood.

***Correct Answer (B, C)**

Analysis of Assessed Standards	
Cluster	Energy
Subcluster	Light
Content	Readiness
Process	
Item Type	Multipart (2 pts)
Stimulus	

Data Analysis

Item	State	Local
Full Credit	38	
No Credit	47	
Partial Credit	15	

Error Analysis
- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.8(C) demonstrate and explain how light travels in a straight line and can be reflected, refracted, or absorbed

OLD **5.6(C)** demonstrate that light travels in a straight line until it strikes an object and is reflected or travels through one medium to another and is refracted

2024 – Q10

What explains why a coin at the bottom of a pool of water seems closer to the surface than it really is?

Ⓐ Light reflects from the surface of the water.

Ⓑ Light reflected from the coin changes direction as it leaves the water.

Ⓒ Light travels only in straight lines after being reflected.

Ⓓ Light reflected from the coin is absorbed by the water.

***Correct Answer (B)**
^highly chosen incorrect answer (A)

Analysis of Assessed Standards

Cluster	Energy
Subcluster	Light
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A	^	
B*	29	
C		
D		

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.8(C) demonstrate and explain how light travels in a straight line and can be reflected, refracted, or absorbed

OLD **5.6(C)** demonstrate that light travels in a straight line until it strikes an object and is reflected or travels through one medium to another and is refracted

2024 – Q24

A student looks at the picture shown.

© Susan Mcarthur Letellier/Dreamstime.com

Which statement **BEST** explains why the trees appear on the surface of the lake?

Ⓐ The trees produce light that reflects off the water.

Ⓑ Light reflects off the trees and reflects off the water.

Ⓒ The trees refract light that then reflects off the water.

Ⓓ Light refracts from the water and is reflected off the trees.

***Correct Answer (B)**
^highly chosen incorrect answer (D)

Analysis of Assessed Standards

Cluster	Energy
Subcluster	Light
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A		
B*	48	
C		
D	^	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

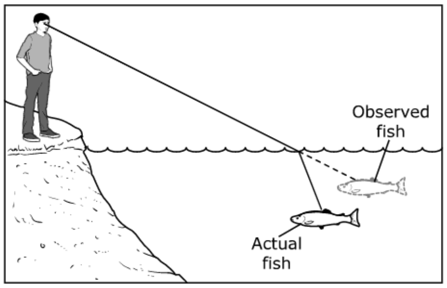
Learning from Mistakes
Instructional Implications

5.8(C) demonstrate and explain how light travels in a straight line and can be reflected, refracted, or absorbed

OLD **5.6(C)** demonstrate that light travels in a straight line until it strikes an object and is reflected or travels through one medium to another and is refracted

2023 – Q7

A person observes a fish in a lake. The position of the fish is shown.



Which statement best explains why the observed fish appears to be closer to the surface than the actual fish?

Ⓐ When light travels from water into air, the light changes direction.

Ⓑ The light bounces off of the water and creates a reflected image of the fish.

Ⓒ When white light passes through the surface of the water, the light separates into colors.

Ⓓ The light enters the water and exits the water at the same angle.

***Correct Answer (A)**

Analysis of Assessed Standards

Cluster	Energy
Subcluster	Light
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A*	30	
B	59	
C	6	
D	5	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

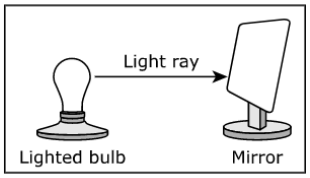
Learning from Mistakes
Instructional Implications

5.8(C) demonstrate and explain how light travels in a straight line and can be reflected, refracted, or absorbed

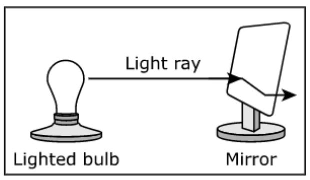
OLD **5.6(C)** demonstrate that light travels in a straight line until it strikes an object and is reflected or travels through one medium to another and is refracted

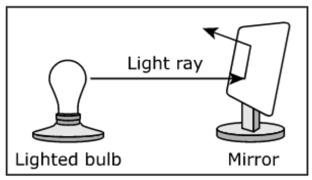
2023 – Q31

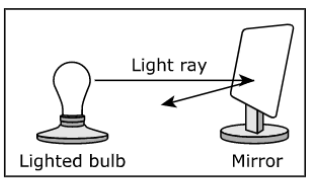
The diagram shows a light ray that is traveling toward a mirror that is tilted upward.

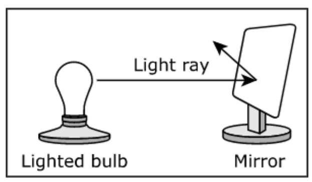


Which drawing best shows how the light ray travels after it reaches the surface of the mirror?

Ⓐ


Ⓒ


Ⓑ


Ⓓ


*Correct Answer (D)

Analysis of Assessed Standards

Cluster	Energy
Subcluster	Light
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A	4	
B	12	
C	9	
D*	74	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.8(C) demonstrate and explain how light travels in a straight line and can be reflected, refracted, or absorbed OLD **5.6(C)** demonstrate that light travels in a straight line until it strikes an object and is reflected or travels through one medium to another and is refracted	**Analysis of Assessed Standards**

2022 – Q3

3 A student is looking directly at a lit nightlight through two different cardboard tubes as shown.

Through which tube, if any, will the light be seen and why?

A The bent tube only because the light bounces off the sides of the tube and travels through the tube to the student's eye

B The straight tube only because the light travels in a straight line directly to the student's eye

C Both tubes because light travels equally well along straight and curved paths

D Neither tube because both tubes absorb all of the light

***Correct Answer (B)**

Cluster	Energy
Subcluster	Light
Content	Readiness
Process	
Stimulus	

Data Analysis

Item	State	Local
A	4	
B*	80	
C	14	
D	2	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.8(C) demonstrate and explain how light travels in a straight line and can be reflected, refracted, or absorbed

OLD **5.6(C)** demonstrate that light travels in a straight line until it strikes an object and is reflected or travels through one medium to another and is refracted

Analysis of Assessed Standards

Cluster	Energy
Subcluster	Light
Content	Readiness
Process	5.4(A)
Stimulus	

2022 – Q19

19 The student is observing part of a plant with a microscope.

Which statement describes a behavior of light in the microscope?

A Light travels through the microscope lens without changing direction.

B Moving in straight lines causes light to increase in brightness.

C Light refracts through the lens of the microscope.

D Moving in straight lines keeps light from reflecting.

***Correct Answer (C)**

Data Analysis

Item	State	Local
A	26	
B	9	
C*	54	
D	11	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.8(C) demonstrate and explain how light travels in a straight line and can be reflected, refracted, or absorbed

OLD **5.6(C)** demonstrate that light travels in a straight line until it strikes an object and is reflected or travels through one medium to another and is refracted

Analysis of Assessed Standards

Cluster	Energy
Subcluster	Light
Content	Readiness
Process	5.3(A)
Stimulus	

2021 – Q3

3 For a class demonstration a student turned off the lights in the classroom. The student then shined light from a flashlight through a hole in a piece of cardboard. The class saw the narrow beam of light continue until another student placed a mirror in the light's path.

The light did not continue past the mirror because —

A light cannot travel very far

B the mirror absorbed all the light

C the light was refracted back to the light source

D light travels in a straight line and cannot go around objects

Data Analysis

Item	State	Local
A	2	
B	12	
C	38	
D*	48	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

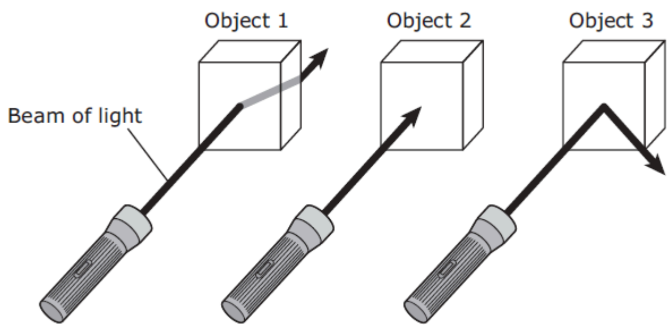
***Correct Answer (D)**

5.8(C) demonstrate and explain how light travels in a straight line and can be reflected, refracted, or absorbed

OLD **5.6(C)** demonstrate that light travels in a straight line until it strikes an object and is reflected or travels through one medium to another and is refracted

2021 – Q24

24 A student uses a flashlight to shine a beam of light on three different objects.



Which table describes what happens to the light beam as it interacts with each object?

F

Object	Light Beam
1	Scattered
2	Absorbed
3	Reflected

H

Object	Light Beam
1	Reflected
2	Absorbed
3	Refracted

G

Object	Light Beam
1	Refracted
2	Absorbed
3	Reflected

J

Object	Light Beam
1	Absorbed
2	Refracted
3	Scattered

***Correct Answer (G)**

Analysis of Assessed Standards

Cluster	Energy
Subcluster	Light
Content	Readiness
Process	5.2(G)
Stimulus	

Data Analysis

Item	State	Local
F	19	
G*	58	
H	12	
J	11	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

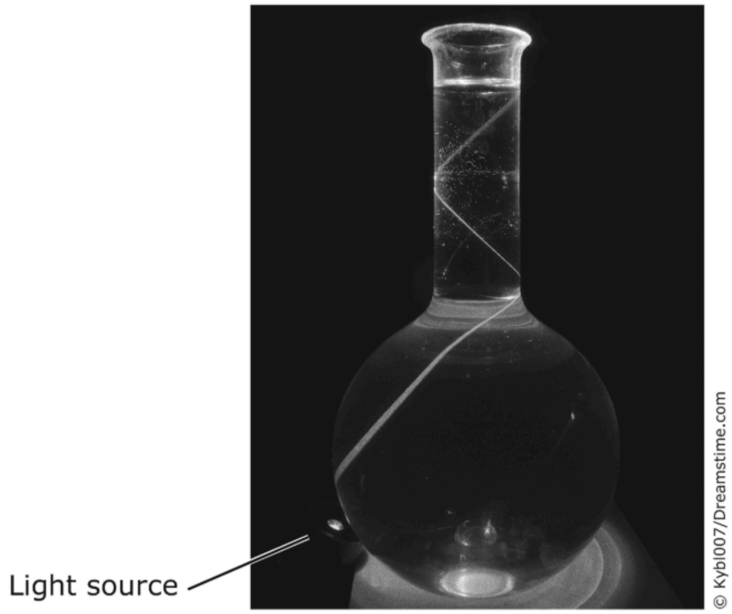
Learning from Mistakes Instructional Implications

5.8(C) demonstrate and explain how light travels in a straight line and can be reflected, refracted, or absorbed

OLD **5.6(C)** demonstrate that light travels in a straight line until it strikes an object and is reflected or travels through one medium to another and is refracted

2019 – Q9

A thin beam of light is shown in this picture.



What does the picture demonstrate about light?

A Light and its reflections travel in straight lines.

B Light cannot reflect from more than one surface.

C Light that goes through water cannot travel in straight lines.

D Light can travel in a circular path.

***Correct Answer (A)**

Analysis of Assessed Standards

Cluster	Energy
Subcluster	Light
Content	Readiness
Process	5.2(C)
Stimulus	

Data Analysis

Item	State	Local
A*	59	
B	4	
C	30	
D	7	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.8(C) demonstrate and explain how light travels in a straight line and can be reflected, refracted, or absorbed

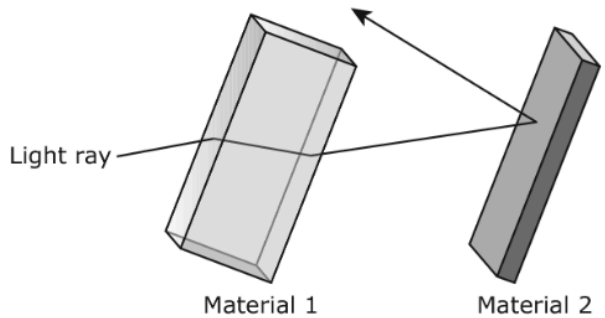
OLD **5.6(C)** demonstrate that light travels in a straight line until it strikes an object and is reflected or travels through one medium to another and is refracted

Analysis of Assessed Standards

Cluster	Energy
Subcluster	Light
Content	Readiness
Process	5.4(A)
Stimulus	

2019 – Q17

The picture shows how a light ray behaves with two different types of materials.



Which table best describes the behavior of the light ray as it encounters the materials?

A

Material 1	Material 2
The light ray is scattered in all directions.	The light ray is refracted.

B

Material 1	Material 2
The light ray is absorbed.	The light ray is reflected.

C

Material 1	Material 2
The light ray is transmitted.	The light ray is refracted.

D

Material 1	Material 2
The light ray is refracted.	The light ray is reflected.

***Correct Answer (D)**

Data Analysis

Item	State	Local
A	6	
B	12	
C	12	
D*	70	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

Space

5.9 Earth and space. The student recognizes patterns among the Sun, Earth, and Moon system and their effects.

IQ Analysis | Investigating the Question	5.9(A)	RC 3

5.9(A) demonstrate that Earth rotates on its axis once approximately every 24 hours and explain how that causes the day/night cycle and the appearance of the Sun moving across the sky, resulting in changes in shadow positions and shapes

2025 – Q4

A teacher places a lamp on a small table. After the lamp is turned on, an office chair is moved near the table. The diagram shows the setup.

How could a student help the teacher demonstrate the day-night cycle of Earth?

Ⓐ The student could sit in the chair and make the chair move around the table.

Ⓑ The student could sit in the chair and cause the chair to make one complete spin.

Ⓒ The student could sit in the chair and then the teacher could move the lamp around the chair.

Ⓓ The student could sit in the chair and then the teacher could walk between the student and the table.

***Correct Answer (B)**
^highly chosen incorrect answer (A)

Analysis of Assessed Standards

Cluster	Space
Subcluster	Earth's Movement
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A	^	
B*	54	
C		
D		

Error Analysis

- ☐ Guessing
- ☐ Mixed Up Concepts
- ☐ Careless Error
- ☐ Stopped Too Early

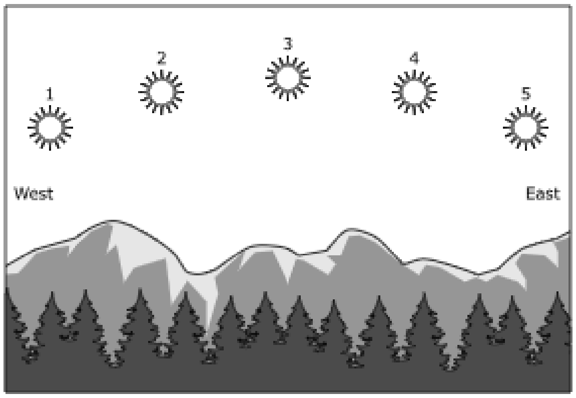
Learning from Mistakes
Instructional Implications

5.9(A) demonstrate that Earth rotates on its axis once approximately every 24 hours and explain how that causes the day/night cycle and the appearance of the Sun moving across the sky, resulting in changes in shadow positions and shapes

2025 – Q13

This question has two parts. First, answer Part A. Then answer Part B.

The diagram shows five positions of the sun in the sky.



A student observes the position of the sun in the sky while outside at noon and again later in the afternoon.

Part A
Which table correctly identifies the **MOST LIKELY** positions at which the student observes the sun?

Ⓐ

Noon	Afternoon
Position 3	Position 2

Ⓑ

Noon	Afternoon
Position 2	Position 4

Ⓒ

Noon	Afternoon
Position 1	Position 5

Ⓓ

Noon	Afternoon
Position 5	Position 3

Part B
Which statement **BEST** explains the answer to Part A?

Ⓐ The sun is visible in the sky during the day and not visible in the sky during the night.

Ⓑ The sun is low in the sky when it is far from Earth and high in the sky when it is close to Earth.

Ⓒ The sun's rotation causes its position to change from east to west throughout the day.

Ⓓ Earth's rotation causes the apparent movement of the sun from east to west.

***Correct Answer (A, D)**

Analysis of Assessed Standards

Cluster	Space
Subcluster	Earth's Movement
Content	Readiness
Process	
Item Type	Multipart (2 pts)
Stimulus	

Data Analysis

Item	State	Local
Full Credit	37	
No Credit	45	
Partial Credit	18	

Error Analysis
- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

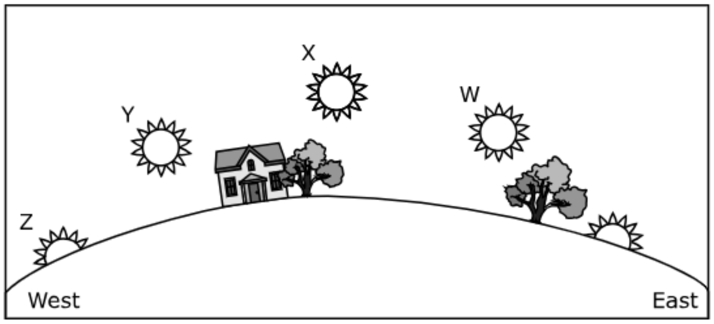
Learning from Mistakes
Instructional Implications

5.9(A) demonstrate that Earth rotates on its axis once approximately every 24 hours and explain how that causes the day/night cycle and the appearance of the Sun moving across the sky, resulting in changes in shadow positions and shapes

OLD **5.8(C)** demonstrate that Earth rotates on its axis once approximately every 24 hours causing the day/night cycle and the apparent movement of the Sun across the sky

2024 – Q8

The diagram shows where the sun appears to be throughout the day.



If the sun will be setting at 6:30 p.m., where will the sun **MOST LIKELY** appear at 3:00 p.m.?

Ⓐ Point W

Ⓑ Point X

Ⓒ Point Y

Ⓓ Point Z

***Correct Answer (C)**
^highly chosen incorrect answer (B)

Analysis of Assessed Standards

Cluster	Space
Subcluster	Earth's Movement
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A		
B	^	
C*	49	
D		

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.9(A) demonstrate that Earth rotates on its axis once approximately every 24 hours and explain how that causes the day/night cycle and the appearance of the Sun moving across the sky, resulting in changes in shadow positions and shapes

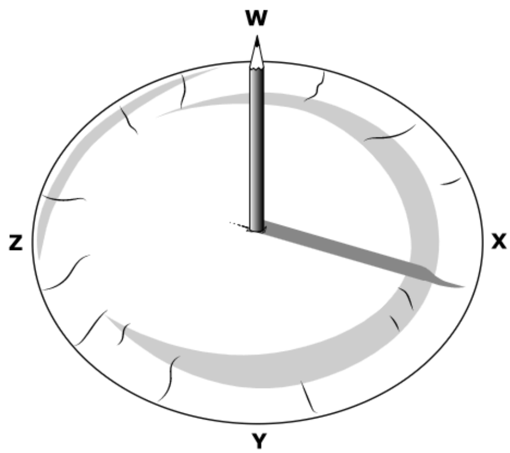
OLD **5.8(C)** demonstrate that Earth rotates on its axis once approximately every 24 hours causing the day/night cycle and the apparent movement of the Sun across the sky

Analysis of Assessed Standards

Cluster	Space
Subcluster	Earth's Movement
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

2023 – Q23

A student makes a sundial. The student takes a picture of the sundial at 3:30 P.M.



Based on the student's sundial, which letter is closest to the position of the sun when the picture was taken?

Ⓐ W

Ⓑ X

Ⓒ Y

Ⓓ Z

Data Analysis

Item	State	Local
A	10	
B	52	
C	4	
D*	34	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

***Correct Answer (D)**

5.9(A) demonstrate that Earth rotates on its axis once approximately every 24 hours and explain how that causes the day/night cycle and the appearance of the Sun moving across the sky, resulting in changes in shadow positions and shapes OLD **5.8(C)** demonstrate that Earth rotates on its axis once approximately every 24 hours causing the day/night cycle and the apparent movement of the Sun across the sky	**Analysis of Assessed Standards**

2022 – Q2

2 Which statement best explains why the sun appears to move across the sky during the day?

F The Earth is closest to the sun in the winter.

G The Earth is revolving around the sun.

H The Earth is tilted at 23.5 degrees.

J The Earth is rotating on its axis.

***Correct Answer (J)**

Cluster	Space
Subcluster	Earth's Movement
Content	Readiness
Process	
Stimulus	

Data Analysis

Item	State	Local
F	1	
G	22	
H	2	
J*	75	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

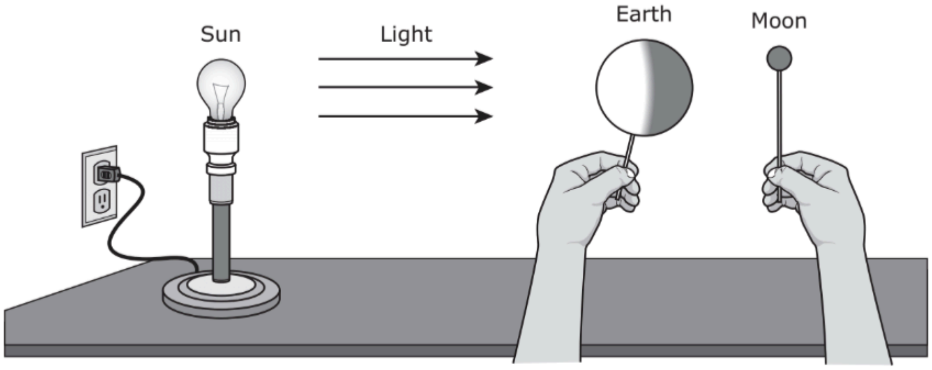
▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.9(A) demonstrate that Earth rotates on its axis once approximately every 24 hours and explain how that causes the day/night cycle and the appearance of the Sun moving across the sky, resulting in changes in shadow positions and shapes OLD **5.8(C)** demonstrate that Earth rotates on its axis once approximately every 24 hours causing the day/night cycle and the apparent movement of the Sun across the sky	**Analysis of Assessed Standards**

2022 – Q26

26 A student makes the model shown with objects representing the sun, Earth, and the moon to use in a class demonstration.



Which action should the student do with the objects to demonstrate a complete day-night cycle of Earth?

F Move the moon around Earth once

G Spin the sun in a circle once

H Move Earth around the sun once

J Spin Earth in a circle once

***Correct Answer (J)**

Cluster	Space
Subcluster	Earth's Movement
Content	Readiness
Process	5.3(C)
Stimulus	

Data Analysis

Item	State	Local
F	14	
G	4	
H	21	
J*	61	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.9(A) demonstrate that Earth rotates on its axis once approximately every 24 hours and explain how that causes the day/night cycle and the appearance of the Sun moving across the sky, resulting in changes in shadow positions and shapes OLD **5.8(C)** demonstrate that Earth rotates on its axis once approximately every 24 hours causing the day/night cycle and the apparent movement of the Sun across the sky	**Analysis of Assessed Standards**

2021 – Q2

2 Earth continuously rotates on its axis while also moving in an orbit. About how much time does it take for Earth to make one complete rotation on its axis?

- **F** 24 hours
- **G** 30 days
- **H** 60 minutes
- **J** 365 days

*__Correct Answer (F)__

Cluster	Space
Subcluster	Earth's Movement
Content	Readiness
Process	
Stimulus	

Data Analysis

Item	State	Local
F*	82	
G	2	
H	1	
J	15	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

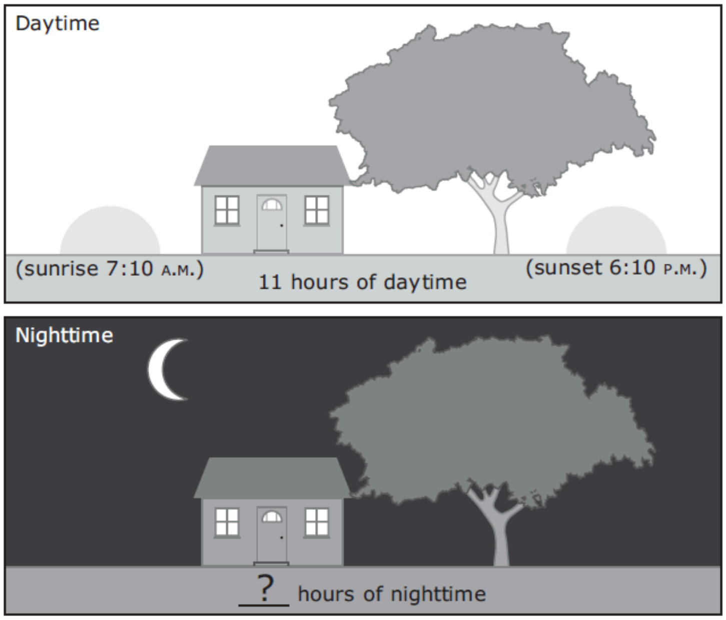
Learning from Mistakes
Instructional Implications

5.9(A) demonstrate that Earth rotates on its axis once approximately every 24 hours and explain how that causes the day/night cycle and the appearance of the Sun moving across the sky, resulting in changes in shadow positions and shapes

OLD **5.8(C)** demonstrate that Earth rotates on its axis once approximately every 24 hours causing the day/night cycle and the apparent movement of the Sun across the sky

2021 – Q23

23 A student drew the following pictures to show the day-night cycle of Earth.



Based on the pictures, how many hours should the student record on the nighttime picture to complete a day-night cycle?

A 11 hours

B 12 hours

C 13 hours

D 24 hours

***Correct Answer (C)**

Analysis of Assessed Standards

Cluster	Space
Subcluster	Earth's Movement
Content	Readiness
Process	
Stimulus	

Data Analysis

Item	State	Local
A	13	
B	15	
C*	58	
D	14	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.9(A) demonstrate that Earth rotates on its axis once approximately every 24 hours and explain how that causes the day/night cycle and the appearance of the Sun moving across the sky, resulting in changes in shadow positions and shapes

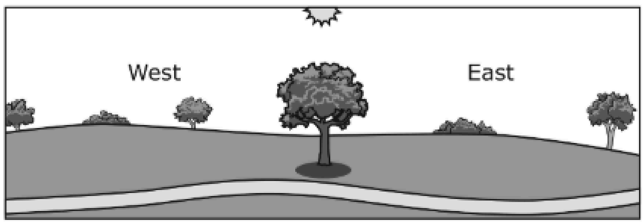
OLD **5.8(C)** demonstrate that Earth rotates on its axis once approximately every 24 hours causing the day/night cycle and the apparent movement of the Sun across the sky

Analysis of Assessed Standards

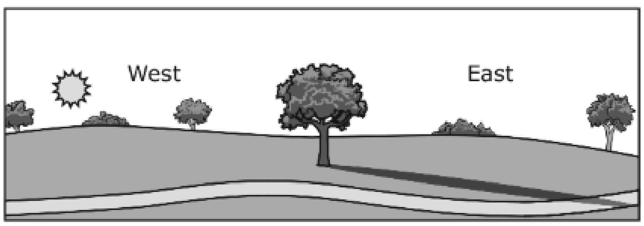
Cluster	Space
Subcluster	Earth's Movement
Content	Readiness
Process	5.2(C)
Stimulus	

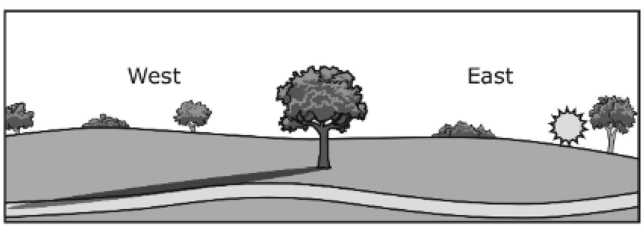
2019 – Q20

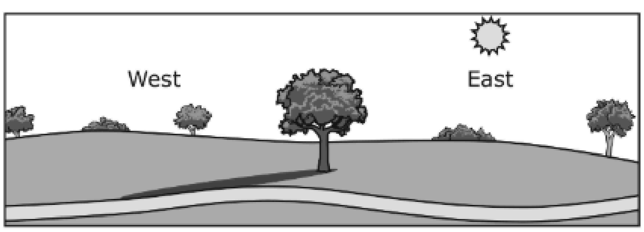
The diagram shows the shadow of a tree in a field at noon on a summer day.

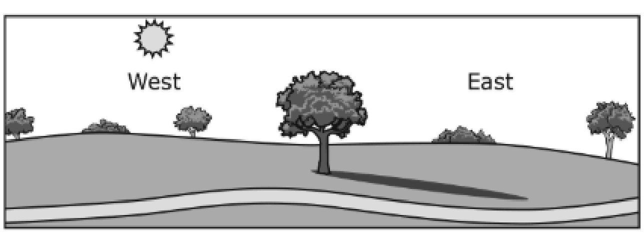


The sun rises at 7:00 A.M. on this day. Which diagram best shows the shadow of the tree at 10:00 A.M. on the same day?

F


G


H


J


Data Analysis

Item	State	Local
F	7	
G	7	
H*	64	
J	23	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

***Correct Answer (H)**

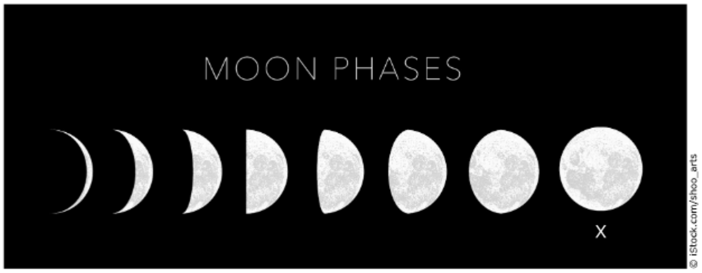
IQ Analysis | Investigating the Question	4.9(B)	RC 3

4.9(B) collect and analyze data to identify sequences and predict patterns of change in the observable appearance of the Moon from Earth

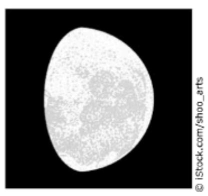
OLD **4.8(C)** collect and analyze data to identify sequences and predict patterns of change in shadows, seasons, and the observable appearance of the Moon over time

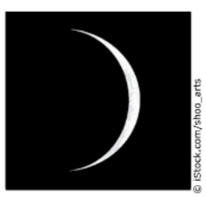
2023 – Q21

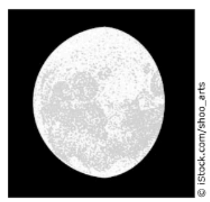
A diagram of some lunar phases is shown.

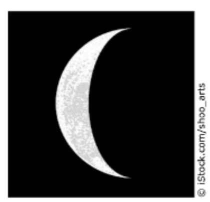


Based on the diagram, which lunar phase will occur directly after X?

Ⓐ


Ⓑ


Ⓒ


Ⓓ


***Correct Answer (C)**

Analysis of Assessed Standards

Cluster	Space
Subcluster	Earth's Movement
Content	Supporting
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A	9	
B	31	
C*	54	
D	6	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

IQ Analysis | Investigating the Question	3.9(B)	RC 3

3.9(B) identify the order of the planets in Earth's solar system in relation to the Sun

OLD **3.8(D)** identify the planets in Earth's solar system and their position in relation to the Sun

2022 – Q9

9 Students make a solar system model on the playground. The school building represents the sun. They label round objects for each planet.

Which planet should they place at the greatest distance from the school building?

A Jupiter

B Neptune

C Mercury

D Saturn

***Correct Answer (B)**

Analysis of Assessed Standards

Cluster	Space
Subcluster	The Solar System
Content	Supporting
Process	5.3(B)
Stimulus	

Data Analysis

Item	State	Local
A	9	
B*	73	
C	10	
D	8	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

3.9(B) identify the order of the planets in Earth's solar system in relation to the Sun

OLD **3.8(D)** identify the planets in Earth's solar system and their position in relation to the Sun

2021 – Q14

14 In the early 1600s, the astronomer Galileo used a telescope to make observations of Mercury and Venus. What are the positions of these planets in relation to the sun?

F Mercury is the planet closest to the sun, and Venus is the second planet from the sun.

G Mercury is the second planet from the sun, and Venus is the third planet from the sun.

H Venus is the planet closest to the sun, and Mercury is the second planet from the sun.

J Mercury is the third planet from the sun, and Venus is the planet closest to the sun.

***Correct Answer (F)**

Analysis of Assessed Standards

Cluster	Space
Subcluster	The Solar System
Content	Supporting
Process	5.3(C)
Stimulus	

Data Analysis

Item	State	Local
F*	72	
G	10	
H	10	
J	7	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

Earth's Processes

5.11 Earth and space. The student knows that there are recognizable patterns and processes on Earth.

IQ Analysis | Investigating the Question	5.10(A)	RC 3

5.10(A) explain how the Sun and the ocean interact in the water cycle and affect weather

OLD **5.8(B)** explain how the Sun and the ocean interact in the water cycle

2021 – Q18

18 Which process will happen when the sun interacts with the ocean?

F Rain will fall only on the land.

G Water will not evaporate at nighttime.

H Salt water will become fresh water vapor.

J Precipitation will happen less often.

***Correct Answer (H)**

Analysis of Assessed Standards

Cluster	Earth's Processes
Subcluster	Water Cycle
Content	Supporting
Process	
Stimulus	

Data Analysis

Item	State	Local
F	5	
G	18	
H*	62	
J	14	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.10(A) explain how the Sun and the ocean interact in the water cycle and affect weather

OLD **5.8(B)** explain how the Sun and the ocean interact in the water cycle

Analysis of Assessed Standards

Cluster	Earth's Processes
Subcluster	Water Cycle
Content	Supporting
Process	
Stimulus	

2019 – Q2

Which of the following explains how water undergoes a change of state in one stage of the water cycle in order to move from the ocean's surface into the air?

F Liquid water that is heated by the sun becomes a gas and rises.

G Gaseous water rises as it is cooled by the ocean.

H Liquid water that is heated by the sun condenses into rain.

J Gaseous water collects in drops of fog above the ocean's surface.

Data Analysis

Item	State	Local
F*	87	
G	2	
H	8	
J	3	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

***Correct Answer (F)**

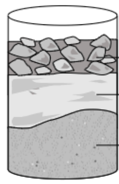
IQ Analysis | Investigating the Question	5.10(B)	RC 3

5.10(B) model and describe the processes that led to the formation of sedimentary rocks and fossil fuels

2025 – Q27

A student uses three sediments to make a cementation model. She layers the sediments in a container as shown.

Incomplete Cementation Model



What should the student do **NEXT** to complete the model of cementation?

- Ⓐ Mix food coloring with each sediment
- Ⓑ Add a heat source above the container
- Ⓒ Shake the container until materials mix
- Ⓓ Add enough glue to fill gaps in materials

***Correct Answer (D)**
^highly chosen incorrect answer (B)

Analysis of Assessed Standards

Cluster	Earth's Processes
Subcluster	Rock Cycle
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A		
B	^	
C		
D*	29	

Error Analysis

- ☐ Guessing
- ☐ Mixed Up Concepts
- ☐ Careless Error
- ☐ Stopped Too Early

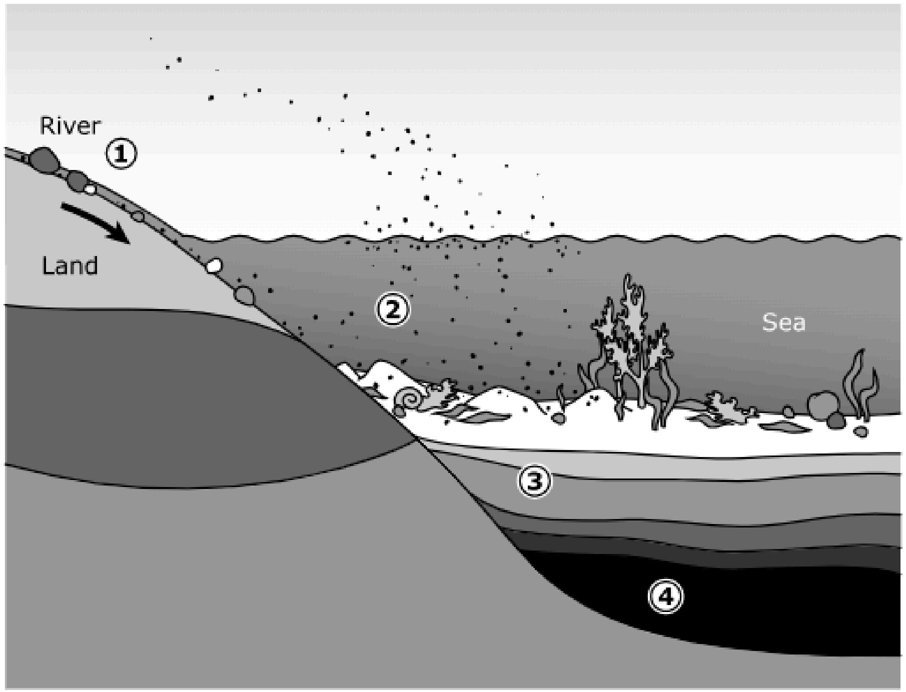
Learning from Mistakes
Instructional Implications

5.10(B) model and describe the processes that led to the formation of sedimentary rocks and fossil fuels

2025 – Q32

This model shows how sedimentary rock is formed.

Model of Sedimentary Rock Formation



What is occurring at Position 2 in the model?

Ⓐ Layers of sediment are being pressed together.

Ⓑ Sediment is being dropped into a new location.

Ⓒ Rocks are being carried to new locations.

Ⓓ Substances are filling in the spaces between particles.

***Correct Answer (B)**
^highly chosen incorrect answer (C)

Analysis of Assessed Standards	
Cluster	Earth's Processes
Subcluster	Rock Cycle
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A		
B*	65	
C	^	
D		

Error Analysis

- ☐ Guessing
- ☐ Mixed Up Concepts
- ☐ Careless Error
- ☐ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.10(B) model and describe the processes that led to the formation of sedimentary rocks and fossil fuels

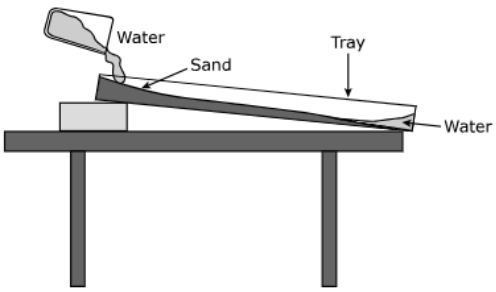
OLD **5.7(A)** explore the processes that led to the formation of sedimentary rocks and fossil fuels

2024 – Q15

A group of students created a model to demonstrate some of the processes involved in the formation of sedimentary rock. The students used this procedure to create the model:

1. Put some sand into a rectangular tray and create a hill at one end with the sand.
2. Raise the end of the tray with the hill of sand by placing a block of wood under it.
3. Create a channel in the sand from the top of the sand hill to the bottom of the sand hill.
4. Pour water into the tray so that it flows through the channel.
5. Observe how the flowing water affects the sand.

The model is shown in the diagram.



Which **TWO** processes of sedimentary rock formation are being modeled **AND** how are they being modeled?

Read the procedure and look at the diagram carefully. Then enter your answer and explanation in the box provided.

Chars 0/475

***Correct Answer (The student must identify that the students are modeling erosion and deposition of sediment AND these processes are being shown by having water erode sand as it travels downhill and deposits the sand at the bottom.)**

Analysis of Assessed Standards

Cluster	Earth's Processes
Subcluster	Rock Cycle
Content	Readiness
Process	
Item Type	Short Constructed Response (2 pts)
Stimulus	

Data Analysis

Item	State	Local
Full Credit	**17**	
No Credit	**59**	
Partial Credit	**23**	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.10(B) model and describe the processes that led to the formation of sedimentary rocks and fossil fuels

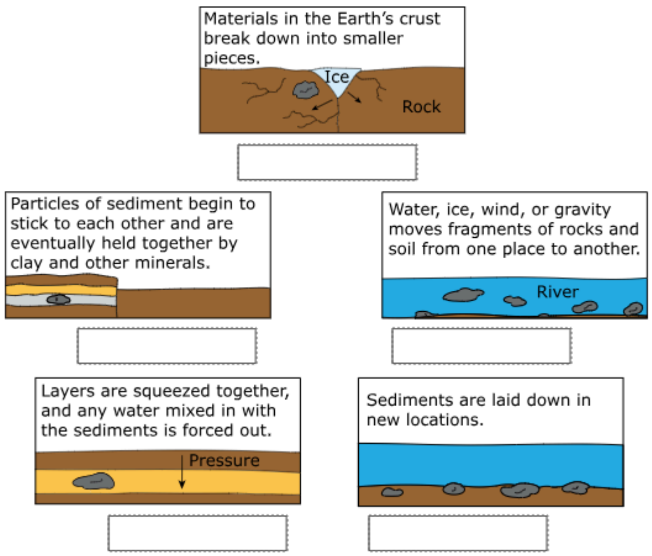
OLD **5.7(A)** explore the processes that led to the formation of sedimentary rocks and fossil fuels

2024 – Q22

Sedimentary rock is formed through several different processes. Match each process with its description.

Move the correct answer to each box.

Compaction	Erosion	Deposition	Weathering	Cementation



***Correct Answer ([clockwise from top] Weathering, Erosion, Deposition, Compaction, Cementation)**

Analysis of Assessed Standards

Cluster	Earth's Processes
Subcluster	Rock Cycle
Content	Readiness
Process	
Item Type	Drag and Drop (2 pts)
Stimulus	

Data Analysis

Item	State	Local
Full Credit	44	
No Credit	32	
Partial Credit	24	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

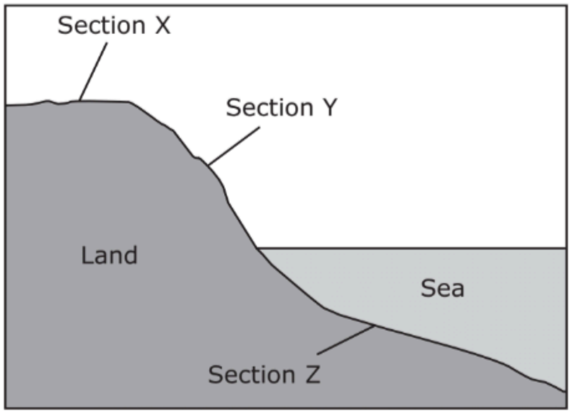
Learning from Mistakes Instructional Implications

5.10(B) model and describe the processes that led to the formation of sedimentary rocks and fossil fuels

OLD **5.7(A)** explore the processes that led to the formation of sedimentary rocks and fossil fuels

2022 – Q6

6 A diagram of different sections of land is shown.



Which action is most likely happening in Section Y in the diagram?

F Wind and rain compacting rock into larger pieces

G Water carrying rocky material to a new location

H Chemicals in water gluing sediments to each other

J Pressure causing layers of sediment to form over time

***Correct Answer (G)**

Analysis of Assessed Standards

Cluster	Earth's Processes
Subcluster	Rock Cycle
Content	Readiness
Process	5.2(D)
Stimulus	

Data Analysis

Item	State	Local
F	17	
G*	40	
H	6	
J	37	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.10(B) model and describe the processes that led to the formation of sedimentary rocks and fossil fuels

OLD **5.7(A)** explore the processes that led to the formation of sedimentary rocks and fossil fuels

2021 – Q26

26 A three-step process is shown.

Deposition → Compaction → Cementation

Which of these are most likely formed by the process shown?

F Glaciers

G Mountains

H Sand dunes

J Sedimentary rocks

***Correct Answer (J)**

Analysis of Assessed Standards

Cluster	Earth's Processes
Subcluster	Rock Cycle
Content	Readiness
Process	
Stimulus	

Data Analysis

Item	State	Local
F	9	
G	12	
H	9	
J*	70	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.10(B) model and describe the processes that led to the formation of sedimentary rocks and fossil fuels

OLD **5.7(A)** explore the processes that led to the formation of sedimentary rocks and fossil fuels

2021 – Q36

36 Sediments are transported at different speeds. Which type of sediment transport is the slowest?

F Transport by rivers

G Transport by winds

H Transport by glaciers

J Transport by ocean currents

***Correct Answer (H)**

Analysis of Assessed Standards

Cluster	Earth's Processes
Subcluster	Rock Cycle
Content	Readiness
Process	
Stimulus	

Data Analysis

Item	State	Local
F	11	
G	16	
H*	63	
J	10	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

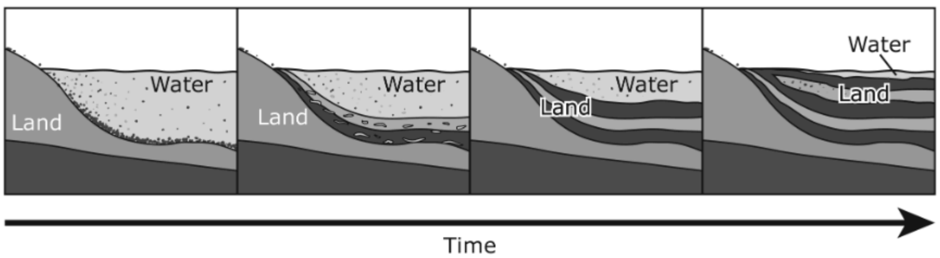
5.10(B) model and describe the processes that led to the formation of sedimentary rocks and fossil fuels

OLD **5.7(A)** explore the processes that led to the formation of sedimentary rocks and fossil fuels

2019 – Q13

The diagram shows parts of a process that occurred over time.

Process Over Time



Which process does this diagram most likely represent?

A The erosion of a coastline

B The deposition of sediments

C The weathering of a mountain

D The formation of a sea

***Correct Answer (B)**

Analysis of Assessed Standards

Cluster	Earth's Processes
Subcluster	Rock Cycle
Content	Readiness
Process	5.3(C)
Stimulus	

Data Analysis

Item	State	Local
A	13	
B*	75	
C	5	
D	6	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

IQ Analysis | Investigating the Question	5.10(C)	RC 3

5.10(C) model and identify how changes to Earth's surface by wind, water, or ice result in the formation of landforms, including deltas, canyons, and sand dunes

2025 – Q6

Which statement **BEST** describes how glaciers change the surface of Earth?

Ⓐ Slow-moving rivers erode the edges of glaciers, forming deltas.

Ⓑ The force of strong winds causes melting ice from a glacier to form canyons.

Ⓒ Glaciers move slowly over the land, weathering and eroding rock layers to form valleys.

Ⓓ The continual melting and freezing of the water in glaciers forms large cracks in the land.

***Correct Answer (C)**
^highly chosen incorrect answer (D)

Analysis of Assessed Standards

Cluster	Earth's Processes
Subcluster	Landforms
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A		
B		
C*	62	
D	^	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.10(C) model and identify how changes to Earth's surface by wind, water, or ice result in the formation of landforms, including deltas, canyons, and sand dunes

2025 – Q25

A group of students took this picture of Box Canyon while hiking in Big Bend National Park.

When looking at the picture, the students notice that Box Canyon has a "V" shape. Which process **MOST LIKELY** formed Box Canyon?

Ⓐ A glacier moving slowly over land

Ⓑ Long-term flooding of rivers and lakes

Ⓒ River water moving quickly over land

Ⓓ Continuous melting and freezing of glaciers

*Correct Answer (C)
^highly chosen incorrect answer (A,B)

Analysis of Assessed Standards

Cluster	Earth's Processes
Subcluster	Landforms
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A	^	
B	^	
C*	45	
D		

Error Analysis

☐ Guessing ☐ Mixed Up Concepts
☐ Careless Error ☐ Stopped Too Early

Learning from Mistakes Instructional Implications

5.10(C) model and identify how changes to Earth's surface by wind, water, or ice result in the formation of landforms, including deltas, canyons, and sand dunes OLD **5.7(B)** recognize how landforms such as deltas, canyons, and sand dunes are the result of changes to Earth's surface by wind, water, or ice	**Analysis of Assessed Standards**

2024 – Q2

A picture of a landform in the desert is shown.

In addition to weathering by rainwater, which process **MOST LIKELY** caused this landform?

Ⓐ Glacier movement

Ⓑ Volcanic eruption

Ⓒ Coastal waves

Ⓓ Wind erosion

*Correct Answer (D)
^highly chosen incorrect answer (C)

Cluster	Earth's Processes
Subcluster	Landforms
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A		
B		
C	^	
D*	58	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

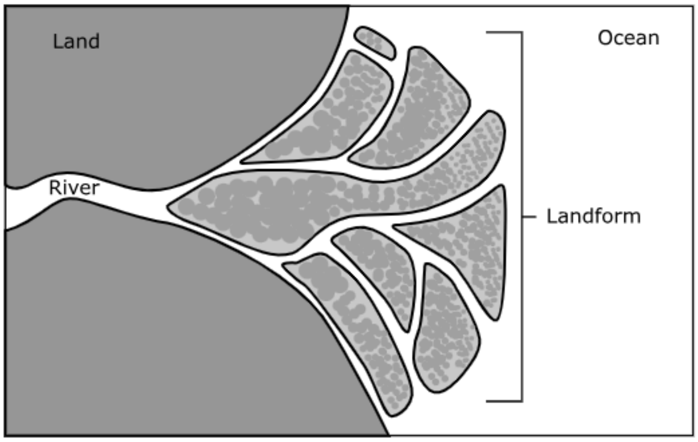
Learning from Mistakes Instructional Implications

5.10(C) model and identify how changes to Earth's surface by wind, water, or ice result in the formation of landforms, including deltas, canyons, and sand dunes

OLD **5.7(B)** recognize how landforms such as deltas, canyons, and sand dunes are the result of changes to Earth's surface by wind, water, or ice

2024 – Q27

Students study a landform that is found at the mouth of a river.



Which statement **BEST** explains the processes that cause this landform?

Ⓐ As ocean waves break on the shore, they deposit sediment from the ocean to the mouth of the river.

Ⓑ As the river water flows into the ocean, it slows down, causing sediments to be deposited at the mouth of the river.

Ⓒ As the river water flows into the ocean, it speeds up, causing sediments to be deposited at the mouth of the river.

Ⓓ As ocean waves break on the shore, they deposit sediment from the shoreline to the mouth of the river.

***Correct Answer (B)**
^highly chosen incorrect answer (C)

Analysis of Assessed Standards

Cluster	Earth's Processes
Subcluster	Landforms
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A		
B*	43	
C	^	
D		

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

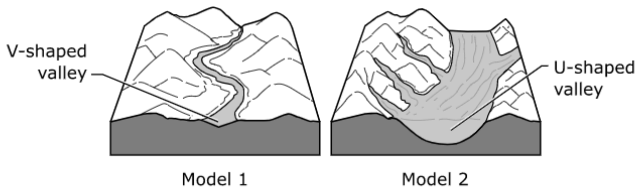
Learning from Mistakes
Instructional Implications

5.10(C) model and identify how changes to Earth's surface by wind, water, or ice result in the formation of landforms, including deltas, canyons, and sand dunes

OLD **5.7(B)** recognize how landforms such as deltas, canyons, and sand dunes are the result of changes to Earth's surface by wind, water, or ice

2023 – Q17

A student builds two models.



Which processes of land formation were the student's models designed to show?

Ⓐ Erosion by fast-moving water and erosion by wind

Ⓑ Valley formation by moving water and valley formation by a glacier

Ⓒ The formation of deltas by slowing water and the formation of oxbow lakes by erosion

Ⓓ Weathering caused by chemical reactions and weathering caused by temperature changes

***Correct Answer (B)**

Analysis of Assessed Standards

Cluster	Earth's Processes
Subcluster	Landforms
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A	17	
B*	67	
C	10	
D	6	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.10(C) model and identify how changes to Earth's surface by wind, water, or ice result in the formation of landforms, including deltas, canyons, and sand dunes

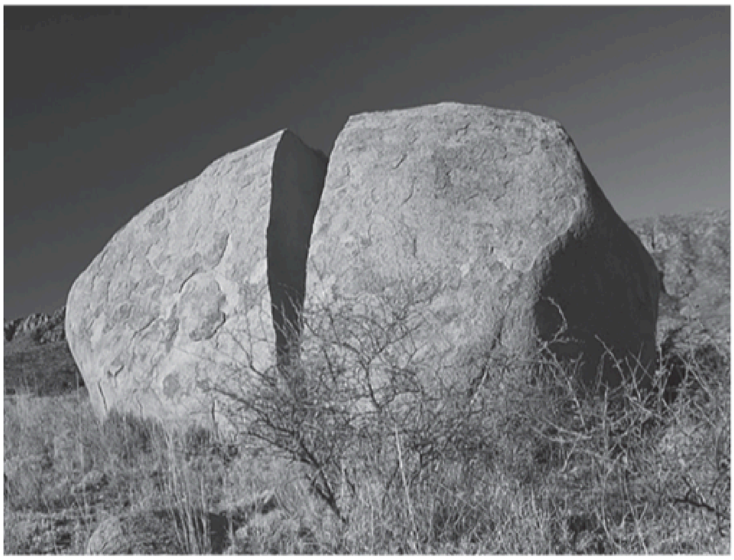
OLD **5.7(B)** recognize how landforms such as deltas, canyons, and sand dunes are the result of changes to Earth's surface by wind, water, or ice

Analysis of Assessed Standards

Cluster	Earth's Processes
Subcluster	Landforms
Content	Readiness
Process	5.2(D)
Stimulus	

2022 – Q14

14 A picture of a rock is shown.



George F. Mobley/National Geographic Creative

Which process most likely caused the crack in the rock?

F Water freezing and thawing in the rock

G Wind blowing particles against the rock

H Water moving and dropping the rock

J Glaciers scraping over the surface of the rock

***Correct Answer (F)**

Data Analysis

Item	State	Local
F*	67	
G	15	
H	7	
J	11	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.10(C) model and identify how changes to Earth's surface by wind, water, or ice result in the formation of landforms, including deltas, canyons, and sand dunes

OLD **5.7(B)** recognize how landforms such as deltas, canyons, and sand dunes are the result of changes to Earth's surface by wind, water, or ice

Analysis of Assessed Standards

Cluster	Earth's Processes
Subcluster	Landforms
Content	Readiness
Process	
Stimulus	

2022 – Q28

28 Which Texas land formation is correctly paired with the force that made the land formation?

F A canyon at Palo Duro Canyon State Park was formed by a river.

G A sand dune at Monahans Sandhills State Park was formed by an earthquake.

H A delta at the end of the Guadalupe River was formed by wind.

J A rock formation with layers on Mustang Island was formed by ice.

Data Analysis

Item	State	Local
F*	63	
G	12	
H	15	
J	10	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

***Correct Answer (F)**

5.10(C) model and identify how changes to Earth's surface by wind, water, or ice result in the formation of landforms, including deltas, canyons, and sand dunes

OLD **5.7(B)** recognize how landforms such as deltas, canyons, and sand dunes are the result of changes to Earth's surface by wind, water, or ice

Analysis of Assessed Standards

Cluster	Earth's Processes
Subcluster	Landforms
Content	Readiness
Process	
Stimulus	

2021 – Q7

7 Canyons and mesas are two landforms found in the western part of the United States. Which statement best describes how canyons and mesas are similar?

A Both were formed by slow-moving glaciers.

B Both were formed by erosion by wind and water.

C Both were part of a mountain that was weathered by ice.

D Both were part of a desert that was reshaped by a flood.

*Correct Answer (B)

Data Analysis

Item	State	Local
A	13	
B*	67	
C	8	
D	12	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.10(C) model and identify how changes to Earth's surface by wind, water, or ice result in the formation of landforms, including deltas, canyons, and sand dunes OLD **5.7(B)** recognize how landforms such as deltas, canyons, and sand dunes are the result of changes to Earth's surface by wind, water, or ice	**Analysis of Assessed Standards**

2021 – Q34

34 Deltas are large landforms found along coastlines. What process forms deltas?

F Cementation of sediments by rivers

G Deposition of sediments by rivers

H Erosion of sediments by ocean waves

J Deposition of sediments by ocean waves

*Correct Answer (G)

Cluster	Earth's Processes
Subcluster	Landforms
Content	Readiness
Process	
Stimulus	

Data Analysis

Item	State	Local
F	10	
G*	51	
H	25	
J	14	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.10(C) model and identify how changes to Earth's surface by wind, water, or ice result in the formation of landforms, including deltas, canyons, and sand dunes OLD **5.7(B)** recognize how landforms such as deltas, canyons, and sand dunes are the result of changes to Earth's surface by wind, water, or ice	**Analysis of Assessed Standards**

2019 – Q8

Which statement correctly describes how a landform is formed?

F A lake is formed when flowing water carves out the sides of a canyon.

G A mountain range is formed when glaciers slowly move across the landscape.

H A delta is formed at the mouth of a river when flowing water slows and deposits sediment.

J A U-shaped valley is formed when winds pick up and move sediment away from the landscape.

***Correct Answer (H)**

Cluster	Earth's Processes
Subcluster	Landforms
Content	Readiness
Process	
Stimulus	

Data Analysis

Item	State	Local
F	10	
G	13	
H*	68	
J	10	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

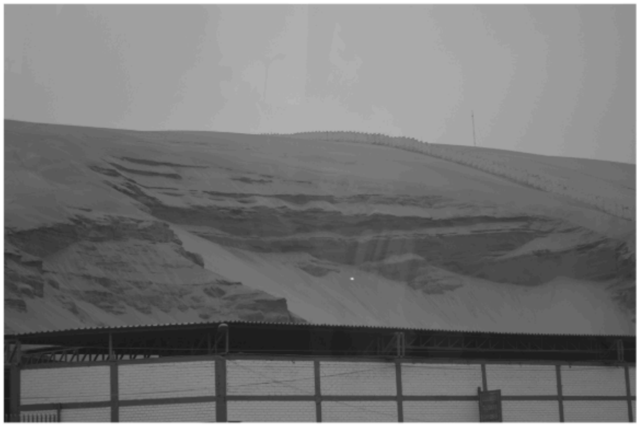
Learning from Mistakes
Instructional Implications

5.10(C) model and identify how changes to Earth's surface by wind, water, or ice result in the formation of landforms, including deltas, canyons, and sand dunes

OLD **5.7(B)** recognize how landforms such as deltas, canyons, and sand dunes are the result of changes to Earth's surface by wind, water, or ice

2019 – Q34

This photograph shows a fence between a sand dune and a road.



What is most likely the main reason this fence was built?

F To stop sand from sliding or blowing onto the road and covering it up

G To prevent a delta from forming at the bottom of the dune

H To keep water off the road

J To prevent marine animals from nesting on the dune

***Correct Answer (F)**

Analysis of Assessed Standards

Cluster	Earth's Processes
Subcluster	Landforms
Content	Readiness
Process	
Stimulus	

Data Analysis

Item	State	Local
F*	77	
G	9	
H	7	
J	7	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

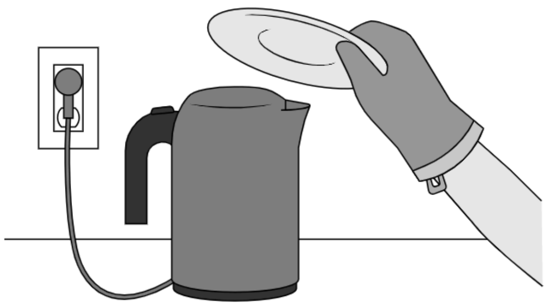
Learning from Mistakes Instructional Implications

IQ Analysis | Investigating the Question	4.10(A)	RC 3

4.10(A) describe and illustrate the continuous movement of water above and on the surface of Earth through the water cycle and explain the role of the Sun as a major source of energy in this process

2025 – Q15

A teacher used an electric kettle filled with water to demonstrate the water cycle to students. The image shows the setup for this demonstration.



Which of these identifies how a stage of the water cycle is represented in the demonstration?

Ⓐ Runoff: Water droplets appeared on the plate.

Ⓑ Condensation: Water droplets fell from the plate.

Ⓒ Evaporation: The amount of water in the kettle decreased.

Ⓓ Precipitation: The water in the kettle began to boil.

***Correct Answer (C)**
^highly chosen incorrect answer (B,D)

Analysis of Assessed Standards

Cluster	Earth's Processes
Subcluster	Water Cycle
Content	Supporting
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A		
B	^	
C*	56	
D	^	

Error Analysis

☐ Guessing ☐ Mixed Up Concepts
☐ Careless Error ☐ Stopped Too Early

Learning from Mistakes
Instructional Implications

4.10(A) describe and illustrate the continuous movement of water above and on the surface of Earth through the water cycle and explain the role of the Sun as a major source of energy in this process

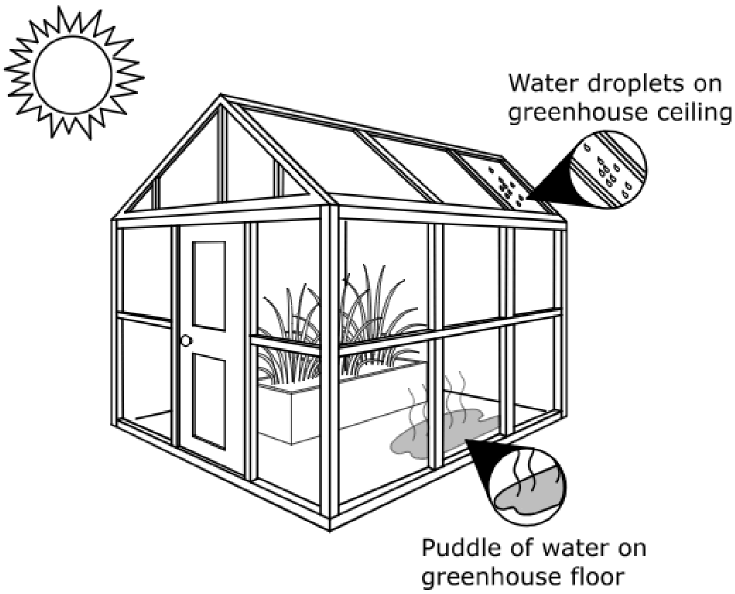
OLD **4.8(B)** describe and illustrate the continuous movement of water above and on the surface of Earth through the water cycle and explain the role of the Sun as a major source of energy in this process

Analysis of Assessed Standards

Cluster	Earth's Processes
Subcluster	Water Cycle
Content	Supporting
Process	
Item Type	Drag and Drop (2 pts)
Stimulus	

2024 – Q13

A diagram of a greenhouse is shown.



Complete the sentences to explain the water in the greenhouse.

Move the correct answer to each box. Not all answers will be used.

evaporation | condensation | solid | liquid | gas

The process of [ ] is occurring on the greenhouse ceiling. This occurs when a [ ] changes to a [ ].

***Correct Answer (condensation, gas, liquid)**

Data Analysis

Item	State	Local
Full Credit	29	
No Credit	63	
Partial Credit	7	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

4.10(A) describe and illustrate the continuous movement of water above and on the surface of Earth through the water cycle and explain the role of the Sun as a major source of energy in this process

OLD **4.8(B)** describe and illustrate the continuous movement of water above and on the surface of Earth through the water cycle and explain the role of the Sun as a major source of energy in this process

Analysis of Assessed Standards

2023 – Q9

To study the water cycle, students add ice cubes to a glass of water. Ten minutes later they observe the glass. On the outside of the glass, students see drops of water.

Which statement explains the students' observations?

Ⓐ Water vapor has become liquid water.

Ⓑ Solid water has become liquid water.

Ⓒ Liquid water has become water vapor.

Ⓓ Solid water has become water vapor.

*Correct Answer (A)

Cluster	Earth's Processes
Subcluster	Water Cycle
Content	Supporting
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A*	29	
B	22	
C	31	
D	18	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

4.10(A) describe and illustrate the continuous movement of water above and on the surface of Earth through the water cycle and explain the role of the Sun as a major source of energy in this process

OLD **4.8(B)** describe and illustrate the continuous movement of water above and on the surface of Earth through the water cycle and explain the role of the Sun as a major source of energy in this process

2022 – Q11

11 A student plans to make a chart that describes each process in the water cycle. Which sentence should the student use to describe the process of condensation?

A Water flows downhill.

B Liquid water turns into water vapor.

C Polar ice turns into liquid water.

D Water vapor collects to form droplets.

***Correct Answer (D)**

Analysis of Assessed Standards

Cluster	Earth's Processes
Subcluster	Water Cycle
Content	Supporting
Process	
Stimulus	

Data Analysis

Item	State	Local
A	6	
B	29	
C	6	
D*	59	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

IQ Analysis | Investigating the Question	4.10(C)	RC 3

4.10(C) differentiate between weather and climate

OLD **5.8(A)** differentiate between weather and climate

2024 – Q21

A student wants to research data about the climate of McAllen, Texas.

Which data should the student research?

Ⓐ The wind speed at different times of the day

Ⓑ The amount of rainfall over several days

Ⓒ The average wind speed and amount of rainfall over one month

Ⓓ The average temperature and amount of rainfall over 30 years

***Correct Answer (D)**
^highly chosen incorrect answer (C)

Analysis of Assessed Standards

Cluster	Earth's Processes
Subcluster	Water Cycle
Content	Supporting
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A		
B		
C	^	
D*	56	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

4.10(C) differentiate between weather and climate

OLD **5.8(A)** differentiate between weather and climate

2019 – Q29

Which table correctly classifies statements about weather and climate?

A

Weather	Climate
Hail can form in thunderstorms that have strong winds.	The high temperature for Tuesday was 28 °C.

B

Weather	Climate
There will be thunderstorms tomorrow in the afternoon.	The month of February has had the coldest temperatures on record for thirty years.

C

Weather	Climate
August has the hottest temperatures each year.	A cold front is expected next week.

D

Weather	Climate
The high temperature for March is usually around 20 °C.	There are clear skies with no chance of rain for the next four days.

***Correct Answer (B)**

Analysis of Assessed Standards

Cluster	Earth's Processes
Subcluster	Water Cycle
Content	Supporting
Process	5.2(G)
Stimulus	

Data Analysis

Item	State	Local
A	12	
B*	74	
C	7	
D	8	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

IQ Analysis | Investigating the Question	3.10(C)	RC 3

3.10(C) model and describe rapid changes in Earth's surface such as volcanic eruptions, earthquakes, and landslides

2025 – Q29

An area of land containing living organisms may rapidly or slowly change to a surface covered with a new rock layer that does not contain any living organisms. Which statement **BEST** explains why this change occurs?

Ⓐ A landslide removes all soil from the area.

Ⓑ A hurricane with high winds causes erosion in the area.

Ⓒ An earthquake changes the land structures of the area.

Ⓓ A volcanic eruption causes lava to flow into the area.

***Correct Answer (D)**
^highly chosen incorrect answer (C)

Analysis of Assessed Standards

Cluster	Earth's Processes
Subcluster	Landforms
Content	Supporting
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A		
B		
C	^	
D*	25	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

3.10(C) model and describe rapid changes in Earth's surface such as volcanic eruptions, earthquakes, and landslides OLD **3.7(B)** investigate rapid changes in Earth's surface such as volcanic eruptions, earthquakes, and landslides	**Analysis of Assessed Standards**

2019 – Q25

Which landform is a result of rapid changes to Earth's surface?

A U-shaped valley

B Limestone cave

C Mountain range

D Volcanic island

***Correct Answer (D)**

Cluster	Earth's Processes
Subcluster	Landforms
Content	Supporting
Process	
Stimulus	

Data Analysis

Item	State	Local
A	23	
B	6	
C	13	
D*	58	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

Earth's Resources

5.11 Earth and space. The student understands how natural resources are important and can be managed.

IQ Analysis | Investigating the Question	5.10(B)	RC 3

5.10(B) model and describe the processes that led to the formation of sedimentary rocks and fossil fuels

2025 – Q10

Which statement **BEST** explains a difference between the formation of sedimentary rock and the formation of fossil fuels?

Ⓐ Heat is needed for the formation of fossil fuels but not for the formation of sedimentary rock.

Ⓑ Cementation occurs in the formation of fossil fuels but not in the formation of sedimentary rock.

Ⓒ Decaying organisms must be present for the formation of sedimentary rock but not for the formation of fossil fuels.

Ⓓ Time is needed for the formation of sedimentary rock but not for the formation of fossil fuels.

***Correct Answer (A)**
^highly chosen incorrect answer (C)

Analysis of Assessed Standards

Cluster	Earth's Resources
Subcluster	Fossil Fuels
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A*	53	
B		
C	^	
D		

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

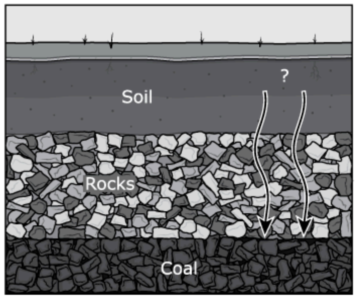
Learning from Mistakes
Instructional Implications

5.10(B) model and describe the processes that led to the formation of sedimentary rocks and fossil fuels

OLD **5.7(A)** explore the processes that led to the formation of sedimentary rocks and fossil fuels

2023 – Q18

The diagram shows how one type of fossil fuel is formed.



How should the arrows in the diagram be labeled?

Ⓐ Pressure and heat

Ⓑ Erosion and deposition

Ⓒ Heat and cementation

Ⓓ Weathering and compaction

***Correct Answer (A)**

Analysis of Assessed Standards

Cluster	Earth's Resources
Subcluster	Fossil Fuels
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A*	57	
B	10	
C	16	
D	18	

Error Analysis

- ☐ Guessing
- ☐ Mixed Up Concepts
- ☐ Careless Error
- ☐ Stopped Too Early

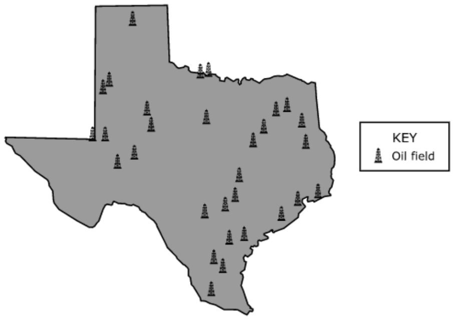
Learning from Mistakes
Instructional Implications

5.10(B) model and describe the processes that led to the formation of sedimentary rocks and fossil fuels

OLD **5.7(A)** explore the processes that led to the formation of sedimentary rocks and fossil fuels

2023 – Q30

The locations of some oil fields in Texas are shown on the map.



Based on the locations of the oil fields, which statement **MOST LIKELY** describes the environmental conditions in Texas millions of years ago?

Ⓐ Texas was once covered by a dry desert.

Ⓑ Texas was once covered by a shallow sea.

Ⓒ Texas was once covered by a flat grassland.

Ⓓ Texas was once covered by a tropical rain forest.

***Correct Answer (B)**

Analysis of Assessed Standards

Cluster	Earth's Resources
Subcluster	Fossil Fuels
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A	20	
B*	35	
C	32	
D	13	

Error Analysis

☐ Guessing ☐ Mixed Up Concepts

☐ Careless Error ☐ Stopped Too Early

Learning from Mistakes Instructional Implications

5.10(B) model and describe the processes that led to the formation of sedimentary rocks and fossil fuels

OLD **5.7(A)** explore the processes that led to the formation of sedimentary rocks and fossil fuels

2022 – Q32

32 Many of America's large oil fields are found underground at the Permian Basin in West Texas. An area of the Permian Basin is shown.

© iStock.com/FerrizFrame

How did these oil fields form?

F Dead plants and animals were buried for millions of years.

G Plants were eaten by consumers that left fossilized remains.

H Heat caused underground rocks to undergo chemical changes.

J Rocks at the surface of Earth melted and then solidified.

***Correct Answer (F)**

Analysis of Assessed Standards

Cluster	Earth's Resources
Subcluster	Fossil Fuels
Content	Readiness
Process	
Stimulus	

Data Analysis

Item	State	Local
F*	62	
G	13	
H	20	
J	6	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.10(B) model and describe the processes that led to the formation of sedimentary rocks and fossil fuels

OLD **5.7(A)** explore the processes that led to the formation of sedimentary rocks and fossil fuels

2019 – Q18

Which of these environments could form coal if the area is buried for a long time?

F 

H 

G 

J 

***Correct Answer (H)**

Analysis of Assessed Standards

Cluster	Earth's Resources
Subcluster	Fossil Fuels
Content	Readiness
Process	
Stimulus	

Data Analysis

Item	State	Local
F	17	
G	9	
H*	45	
J	28	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

IQ Analysis | Investigating the Question	4.11(A)	RC 3

4.11(A) identify and explain advantages and disadvantages of using Earth's renewable and nonrenewable natural resources such as wind, water, sunlight, plants, animals, coal, oil, and natural gas

OLD **4.7(C)** identify and classify Earth's renewable resources, including air, plants, water, and animals; and nonrenewable resources, including coal, oil, and natural gas; and the importance of conservation

Analysis of Assessed Standards

Cluster	Earth's Resources
Subcluster	Resource Management
Content	Supporting
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

2023 – Q11

Students make a table of renewable and nonrenewable resources. Which table correctly identifies resources as renewable or nonrenewable?

Ⓐ

Renewable	Nonrenewable
Water	Coal
Wood	Solar

Ⓑ

Renewable	Nonrenewable
Wind	Minerals
Water	Oil

Ⓒ

Renewable	Nonrenewable
Coal	Water
Natural Gas	Minerals

Ⓓ

Renewable	Nonrenewable
Wood	Wind
Water	Oil

Data Analysis

Item	State	Local
A	20	
B*	61	
C	9	
D	10	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

***Correct Answer (B)**

4.11(A) identify and explain advantages and disadvantages of using Earth's renewable and nonrenewable natural resources such as wind, water, sunlight, plants, animals, coal, oil, and natural gas OLD **4.7(C)** identify and classify Earth's renewable resources, including air, plants, water, and animals; and nonrenewable resources, including coal, oil, and natural gas; and the importance of conservation	**Analysis of Assessed Standards**

2022 – Q20

20 A student lists resources that can be used to produce electricity. The list is shown.

- Wind
- Coal
- Natural gas
- Water power
- Petroleum

Which of the resources are renewable resources?

F Wind, water power

G Wind, coal, and natural gas

H Coal, water power, and petroleum

J Coal, natural gas, water power, and petroleum

***Correct Answer (F)**

Cluster	Earth's Resources
Subcluster	Resource Management
Content	Supporting
Process	
Stimulus	

Data Analysis

Item	State	Local
F*	68	
G	13	
H	9	
J	10	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

4.11(A) identify and explain advantages and disadvantages of using Earth's renewable and nonrenewable natural resources such as wind, water, sunlight, plants, animals, coal, oil, and natural gas

OLD **4.7(C)** identify and classify Earth's renewable resources, including air, plants, water, and animals; and nonrenewable resources, including coal, oil, and natural gas; and the importance of conservation

2021 – Q31

31 Students list resources that are used to heat buildings.

- Wood
- Coal
- Natural gas
- Petroleum
- Solar energy

Which of these tables correctly classifies the resources?

A

Renewable	Nonrenewable
Solar energy	Petroleum
Wood	Natural gas
	Coal

B

Renewable	Nonrenewable
Solar energy	Coal
Natural gas	Petroleum
	Wood

C

Renewable	Nonrenewable
Wood	Solar energy
Coal	Petroleum
	Natural gas

D

Renewable	Nonrenewable
Petroleum	Solar energy
Natural gas	Wood
Coal	

***Correct Answer (A)**

Analysis of Assessed Standards

Cluster	Earth's Resources
Subcluster	Resource Management
Content	Supporting
Process	
Stimulus	

Data Analysis

Item	State	Local
A*	60	
B	17	
C	14	
D	9	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

4.11(A) identify and explain advantages and disadvantages of using Earth's renewable and nonrenewable natural resources such as wind, water, sunlight, plants, animals, coal, oil, and natural gas

OLD **4.7(C)** identify and classify Earth's renewable resources, including air, plants, water, and animals; and nonrenewable resources, including coal, oil, and natural gas; and the importance of conservation

Analysis of Assessed Standards

Cluster	Earth's Resources
Subcluster	Resource Management
Content	Supporting
Process	
Stimulus	

2019 – Q28

A student read an article about natural resources and listed some resources in the chart shown.

Natural Resources

Coal	Air	Tree
Petroleum	Wheat	Natural gas
Sheep	Lake	Fish

Which answer choice best classifies the natural resources in the chart?

F Renewable: coal, air, tree
Nonrenewable: petroleum, wheat, natural gas, sheep, lake, fish

G Renewable: coal, petroleum, natural gas, lake
Nonrenewable: air, tree, wheat, sheep, fish

H Renewable: tree, wheat, sheep, fish
Nonrenewable: coal, air, petroleum, natural gas, lake

J Renewable: air, tree, wheat, sheep, lake, fish
Nonrenewable: coal, petroleum, natural gas

***Correct Answer (J)**

Data Analysis

Item	State	Local
F	9	
G	8	
H	15	
J*	68	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

Organisms and Environments

5.12 Organisms and environments. The student describes patterns, cycles, systems, and relationships within environments.

IQ Analysis | Investigating the Question	5.12(A)	RC 4

5.12(A) observe and describe how a variety of organisms survive by interacting with biotic and abiotic factors in a healthy ecosystem

2025 – Q9

Which statement describes one way trees are dependent on nonliving parts of the environment?

Ⓐ Many birds build nests in trees.

Ⓑ Different organisms rely on trees to provide food.

Ⓒ Some trees need fire to release their seeds.

Ⓓ Trees provide shade for plants that grow near them.

***Correct Answer (C)**
^highly chosen incorrect answer (B)

Analysis of Assessed Standards

Cluster	Organisms and Environments
Subcluster	Biotic and Abiotic Factors
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A		
B	^	
C*	48	
D		

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.12(A) observe and describe how a variety of organisms survive by interacting with biotic and abiotic factors in a healthy ecosystem

2025 – Q26

The Egyptian plover and the Nile crocodile can both be found along the Nile River. The Egyptian plover is a bird that cleans the Nile crocodile's teeth by picking out and eating meat that is stuck between the crocodile's teeth. The Nile crocodile does not eat the Egyptian plover.

Which statement describes a relationship between a living and a nonliving component of this scenario?

Ⓐ The Nile crocodile lives along the Nile River.

Ⓑ The Nile crocodile is a carnivore.

Ⓒ The Egyptian plover preys on the Nile crocodile.

Ⓓ The Egyptian plover lands on the Nile crocodile.

*Correct Answer (A)
^highly chosen incorrect answer (B)

Analysis of Assessed Standards

Cluster	Organisms and Environments
Subcluster	Biotic and Abiotic Factors
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A*	57	
B	^	
C		
D		

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.12(A) observe and describe how a variety of organisms survive by interacting with biotic and abiotic factors in a healthy ecosystem

2025 – Q30

The Texas paintbrush is a native Texas wildflower.

The roots of this plant can pull needed water and nutrients from the root systems of nearby plants.

Which statement describes another way that this interaction between the Texas paintbrush and other plants helps the Texas paintbrush survive in its environment?

Ⓐ The Texas paintbrush is protected from animals that feed on plants.

Ⓑ The Texas paintbrush slows the growth of nearby plants so that it has plenty of room to grow.

Ⓒ The Texas paintbrush grows during different times of the year than nearby plants do.

Ⓓ The Texas paintbrush increases in its attractiveness to bees and other pollinators that help it reproduce.

***Correct Answer (B)**
^highly chosen incorrect answer (D)

Analysis of Assessed Standards

Cluster	Organisms and Environments
Subcluster	Biotic and Abiotic Factors
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A		
B*	43	
C		
D	^	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.12(A) observe and describe how a variety of organisms survive by interacting with biotic and abiotic factors in a healthy ecosystem

OLD **5.9(A)** observe the way organisms live and survive in their ecosystem by interacting with the living and nonliving components

2024 – Q20

Which pair of activities are examples of bumblebees interacting with living parts in their environment?

Ⓐ Collecting pollen from flowers
Building a nest on the side of a shed

Ⓑ Drinking water from a water trough
Staying underground during cold months

Ⓒ Passing stored nectar from one worker to another worker
Forming a nest in a hole in a tree

Ⓓ Queen bee laying eggs
Worker bees using wax to make a hive

***Correct Answer (C)**
^highly chosen incorrect answer (A)

Analysis of Assessed Standards

Cluster	Organisms and Environments
Subcluster	Biotic and Abiotic Factors
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A	^	
B		
C*	52	
D		

Error Analysis

▢ Guessing ▢ Mixed Up Concepts
▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.12(A) observe and describe how a variety of organisms survive by interacting with biotic and abiotic factors in a healthy ecosystem

OLD **5.9(A)** observe the way organisms live and survive in their ecosystem by interacting with the living and nonliving components

Analysis of Assessed Standards

2023 – Q10

Which sentence describes the interaction of a white-tailed deer with both living and nonliving parts of its ecosystem?

Ⓐ The white-tailed deer hides behind a rock and uses the heat from the sun to stay warm.

Ⓑ The white-tailed deer protects its fawn from predators and shows the fawn how to find food.

Ⓒ The white-tailed deer rubs its antlers on a tree and eats leaves from the tree.

Ⓓ The white-tailed deer eats grass in a field and drinks water from a pond.

*Correct Answer (D)

Cluster	Organisms and Environments
Subcluster	Biotic and Abiotic Factors
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A	17	
B	12	
C	12	
D*	58	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.12(A) observe and describe how a variety of organisms survive by interacting with biotic and abiotic factors in a healthy ecosystem

OLD **5.9(A)** observe the way organisms live and survive in their ecosystem by interacting with the living and nonliving components

Analysis of Assessed Standards

Cluster	Organisms and Environments
Subcluster	Biotic and Abiotic Factors
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

2023 – Q32

A teacher uses an aquarium to make a habitat for a frog. A list of student observations of the frog is shown.

Student Observations

1. Hiding in the leaves of a plant
2. Eating a mealworm
3. Swimming through the water
4. Burrowing in the soil
5. Catching a small cricket
6. Hopping over a rock

Which student observations show the frog interacting with nonliving elements in its environment?

Ⓐ 1 and 4

Ⓑ 2 and 5

Ⓒ 3 and 6

Ⓓ 6 and 1

***Correct Answer (C)**

Data Analysis

Item	State	Local
A	10	
B	6	
C*	75	
D	9	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.12(A) observe and describe how a variety of organisms survive by interacting with biotic and abiotic factors in a healthy ecosystem OLD **5.9(A)** observe the way organisms live and survive in their ecosystem by interacting with the living and nonliving components	**Analysis of Assessed Standards**

2022 – Q12

12 Students study barracudas. They gather some observations of barracudas. A barracuda is shown.

Barracuda

Which observation describes barracudas interacting with the living elements of their ecosystem?

F Barracudas can travel quickly using surface ocean currents.

G Barracudas live around hard structures such as oil rigs and jetties.

H Barracudas are predators of other fish.

J Barracudas tend to live in warm waters.

***Correct Answer (H)**

Cluster	Organisms and Environments
Subcluster	Biotic and Abiotic Factors
Content	Readiness
Process	
Stimulus	

Data Analysis

Item	State	Local
F	16	
G	10	
H*	67	
J	7	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.12(A) observe and describe how a variety of organisms survive by interacting with biotic and abiotic factors in a healthy ecosystem OLD **5.9(A)** observe the way organisms live and survive in their ecosystem by interacting with the living and nonliving components	**Analysis of Assessed Standards**

2022 – Q17

17 Two box turtles live in an area in a zoo's reptile house. Zoo visitors made a list of observations of the turtles. The list is shown.

1. Sliding into a small pond
2. Eating a strawberry
3. Digging a hole in the sandy soil
4. Climbing onto a flat rock
5. Holding an earthworm in its mouth
6. Walking across the area

Which observations best describe how box turtles interact with living parts of their environment?

A Observation 4 and observation 6

B Observation 2 and observation 5

C Observation 1 and observation 5

D Observation 3 and observation 4

***Correct Answer (B)**

Cluster	Organisms and Environments
Subcluster	Biotic and Abiotic Factors
Content	Readiness
Process	
Stimulus	

Data Analysis

Item	State	Local
A	9	
B*	62	
C	16	
D	13	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.12(A) observe and describe how a variety of organisms survive by interacting with biotic and abiotic factors in a healthy ecosystem

OLD **5.9(A)** observe the way organisms live and survive in their ecosystem by interacting with the living and nonliving components

Analysis of Assessed Standards

Cluster	Organisms and Environments
Subcluster	Biotic and Abiotic Factors
Content	Readiness
Process	
Stimulus	

2021 – Q22

22 The chart lists plants and animals interacting with parts of an environment.

1. A hummingbird dips its beak inside a flower.
2. A lizard burrows into the sand to stay cool.
3. A fish absorbs oxygen through its gills.
4. A cactus wren eats seeds from a cactus fruit.
5. A sea star clings to a rock in a tidal pool.
6. A bear scratches its back against a tree.

Which statements describe an animal interacting with a living part of the environment?

F Statements 1, 4, and 6 only

G Statements 1, 3, and 5 only

H Statements 2, 5, and 6 only

J Statements 2, 3, and 4 only

*Correct Answer (F)

Data Analysis

Item	State	Local
F*	72	
G	10	
H	11	
J	7	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.12(A) observe and describe how a variety of organisms survive by interacting with biotic and abiotic factors in a healthy ecosystem

OLD **5.9(A)** observe the way organisms live and survive in their ecosystem by interacting with the living and nonliving components

Analysis of Assessed Standards

Cluster	Organisms and Environments
Subcluster	Biotic and Abiotic Factors
Content	Readiness
Process	5.2(D)
Stimulus	

Data Analysis

Item	State	Local
A	7	
B	11	
C	6	
D*	77	

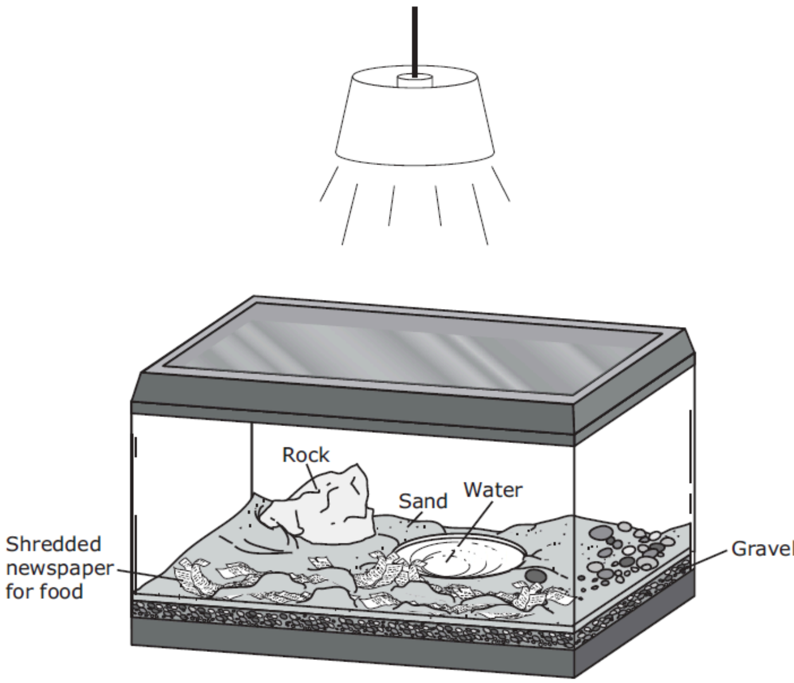
Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

2021 – Q33

33 A student's model of a closed ecosystem is shown. The student plans to add land snails to the model in order to show how the snails interact with different parts of the ecosystem.



What should the student add to the model so that the snails will survive?

A Fish to provide food

B Sticks to provide a place to hide

C Deep water to provide carbon dioxide

D Living plants to provide oxygen

***Correct Answer (D)**

5.12(A) observe and describe how a variety of organisms survive by interacting with biotic and abiotic factors in a healthy ecosystem OLD **5.9(A)** observe the way organisms live and survive in their ecosystem by interacting with the living and nonliving components	**Analysis of Assessed Standards**

2019 – Q19

A zookeeper sets up a habitat for a certain species of rabbit. Which answer choice describes the rabbit as it interacts with two nonliving parts of the rabbit's habitat?

A The rabbit breathing air while running down a rocky path

B The rabbit licking its fur while feeding its young

C The rabbit hiding under plants from loud noises

D The rabbit nibbling on vegetation inside the habitat

***Correct Answer (A)**

Cluster	Organisms and Environments
Subcluster	Biotic and Abiotic Factors
Content	Readiness
Process	5.3(C)
Stimulus	

Data Analysis

Item	State	Local
A*	73	
B	5	
C	9	
D	12	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.12(A) observe and describe how a variety of organisms survive by interacting with biotic and abiotic factors in a healthy ecosystem

OLD **5.9(A)** observe the way organisms live and survive in their ecosystem by interacting with the living and nonliving components

2019 – Q27

A student observed and recorded some activities in an aquarium.

Observations

1. A fish eats flakes of fish food dropped into the aquarium.
2. A snail crawls over colored rocks at the bottom of the aquarium.
3. A fish eats leaves from a plant in the aquarium.
4. A snail lays eggs in a corner of the aquarium.
5. A fish swims through air bubbles being pumped into the aquarium.
6. A snail moves up a wall of the aquarium.

An interaction between two living parts of the environment is represented by —

A Observations 1 and 2

B Observation 3

C Observations 4 and 5

D Observation 6

***Correct Answer (B)**

Analysis of Assessed Standards

Cluster	Organisms and Environments
Subcluster	Biotic and Abiotic Factors
Content	Readiness
Process	5.2(D)
Stimulus	

Data Analysis

Item	State	Local
A	10	
B*	73	
C	13	
D	3	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

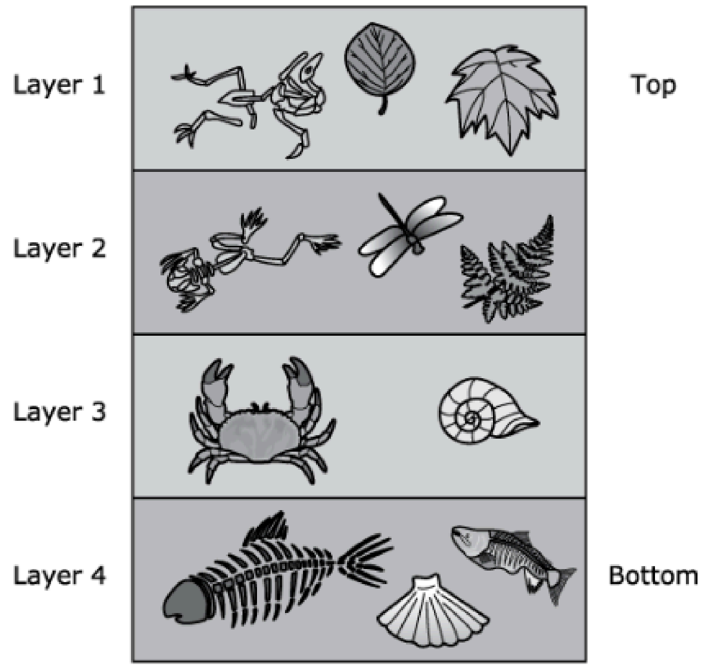
Learning from Mistakes
Instructional Implications

IQ Analysis | Investigating the Question	3.12(D)	RC 4

3.12(D) identify fossils as evidence of past living organisms and environments, including common Texas fossils

2025 – Q1

This drawing shows fossils in different layers of soil.



Based on the fossils, which statement **BEST** describes one of the layers?

Ⓐ Layer 1 was a desert.

Ⓑ Layer 2 was an Arctic tundra.

Ⓒ Layer 3 was an ocean.

Ⓓ Layer 4 was a forest.

***Correct Answer (C)**
^highly chosen incorrect answer (B)

Analysis of Assessed Standards

Cluster	Organisms and Environments
Subcluster	Fossils
Content	Supporting
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A		
B	^	
C*	89	
D		

Error Analysis

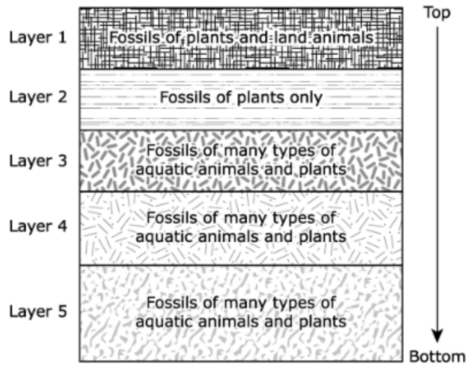
- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

3.12(D) identify fossils as evidence of past living organisms and environments, including common Texas fossils

2025 – Q24

Scientists examine the diagram shown, which has information about the fossils found in different layers of rock.



The scientists believe that a sudden change in the environment at one point led to the disappearance of all the animals in the area.

Which statements are supported by the information in the diagram?

Select **TWO** correct answers.

- ☐ Large trees most likely grew in Layers 4 and 5.
- ☐ At one time the entire area was covered by an ocean.
- ☐ The formation of Layer 4 occurred before the formation of Layer 5.
- ☐ Layer 1 supported only large populations of aquatic plants.
- ☐ The disappearance of the animals occurred before the formation of Layer 1.

***Correct Answer (B, E)**

Analysis of Assessed Standards

Cluster	Organisms and Environments
Subcluster	Fossils
Content	Supporting
Process	
Item Type	Multiselect (2 pts)
Stimulus	

Data Analysis

Item	State	Local
Full Credit	41	
No Credit	16	
Partial Credit	43	

Error Analysis

☐ Guessing ☐ Mixed Up Concepts

☐ Careless Error ☐ Stopped Too Early

Learning from Mistakes Instructional Implications

3.12(D) identify fossils as evidence of past living organisms and environments, including common Texas fossils

OLD **5.9(D)** identify fossils as evidence of past living organisms and the nature of the environments at the time using models

Analysis of Assessed Standards

Cluster	Organisms and Environments
Subcluster	Fossils
Content	Supporting
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

2024 – Q6

Students found a rock with fossils in a dry streambed. Which question can **MOST LIKELY** be answered by identifying the fossils in the rock?

Ⓐ How deep was the water millions of years ago where the fossils are located?

Ⓑ What types of organisms now live in the stream when it rains hard and fills with water?

Ⓒ What was the environment like millions of years ago in the area of the dry streambed?

Ⓓ How many different organisms are now living in the area of the dry streambed?

Data Analysis

Item	State	Local
A	^	
B		
C*	67	
D		

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

***Correct Answer (C)**
^highly chosen incorrect answer (A)

3.12(D) identify fossils as evidence of past living organisms and environments, including common Texas fossils

OLD **5.9(D)** identify fossils as evidence of past living organisms and the nature of the environments at the time using models

Analysis of Assessed Standards

Cluster	Organisms and Environments
Subcluster	Fossils
Content	Supporting
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

2023 – Q22

Researchers found seashell fossils, including those from mollusks and coral, in a rock formation high in the Rocky Mountains.

Which statement **BEST** explains why seashell fossils can be found on some mountains?

Ⓐ The rocks formed under water.

Ⓑ Rivers carried the rocks up the mountains.

Ⓒ The area was a sand dune that hardened as it was covered by water.

Ⓓ The mountains split to form canyons that contained organisms with shells.

*Correct Answer (A)

Data Analysis

Item	State	Local
A*	36	
B	25	
C	19	
D	20	

Error Analysis

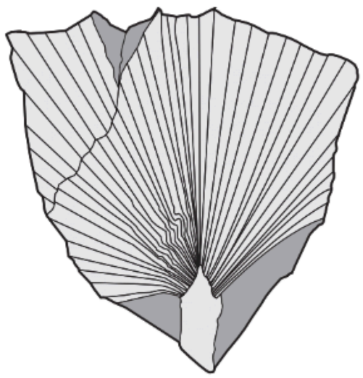
- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

3.12(D) identify fossils as evidence of past living organisms and environments, including common Texas fossils OLD **5.9(D)** identify fossils as evidence of past living organisms and the nature of the environments at the time using models	**Analysis of Assessed Standards**

2022 – Q27

27 Scientists have uncovered palm plant fossils in Alaska, the northern-most state of the United States. Modern-day palm plants grow in tropical climates. A picture of a palm fossil that measures almost one meter across is shown.



This discovery suggests to scientists that this area in Alaska once was —

A covered with an ocean

B warmer than it is now

C populated by polar bears

D changed by earthquakes

***Correct Answer (B)**

Cluster	Organisms and Environments
Subcluster	Fossils
Content	Supporting
Process	5.2(D)
Stimulus	

Data Analysis

Item	State	Local
A	33	
B*	50	
C	5	
D	13	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

3.12(D) identify fossils as evidence of past living organisms and environments, including common Texas fossils

OLD **5.9(D)** identify fossils as evidence of past living organisms and the nature of the environments at the time using models

Analysis of Assessed Standards

Cluster	Organisms and Environments
Subcluster	Fossils
Content	Supporting
Process	5.2(B)
Stimulus	

2021 – Q13

13 A dinosaur fossil is shown.

Which question can scientists most likely answer from studying the fossil of the dinosaur?

A What was the pattern on the skin of the dinosaur?

B What was the type of food the dinosaur ate?

C How fast was the heart of the dinosaur beating?

D How many eggs were in the nest of the dinosaur?

***Correct Answer (B)**

Data Analysis

Item	State	Local
A	32	
B*	61	
C	4	
D	3	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

3.12(D) identify fossils as evidence of past living organisms and environments, including common Texas fossils

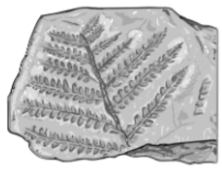
OLD **5.9(D)** identify fossils as evidence of past living organisms and the nature of the environments at the time using models

Analysis of Assessed Standards

Cluster	Organisms and Environments
Subcluster	Fossils
Content	Supporting
Process	5.3(C)
Stimulus	

2019 – Q23

A scientist finds the plant fossil shown.



Which question can the scientist most likely answer by examining this fossil?

A What was the average monthly rainfall in the area?

B How much water was absorbed by the roots of the plant?

C How much oxygen was in the atmosphere surrounding the plant?

D What was the environment like in the area when the plant was alive?

***Correct Answer (D)**

Data Analysis

Item	State	Local
A	6	
B	9	
C	5	
D*	80	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

Structure, Function, and Survival

5.13 Organisms and environments. The student knows that organisms undergo similar life processes and have structures and behaviors that help them survive within their environments.

IQ Analysis | Investigating the Question	5.13(A)	RC 4

5.13(A) analyze the structures and functions of different species to identify how organisms survive in the same environment

2025 – Q5

Adaptations of three animals are shown in the table.

	Fennec Fox	Badger	Hare
Image			
Adaptations	• Large ears for cooling • Light color fur for camouflage	• Sharp claws to dig burrows • Flattens body to get through soil	• Thick fur • Small body to save heat

Based on the table, in which environment is each animal **MOST LIKELY** found?

Move the correct answer to each box in the table.

Arctic Desert Prairie Ocean Wetlands

Animal	Environment
Fennec fox	
Badger	
Hare	

***Correct Answer (Fennec fox = Desert; Badger = Prairie; Hare = Arctic)**

Analysis of Assessed Standards	
Cluster	Structure, Function, and Survival
Subcluster	Structure and Function
Content	Readiness
Process	
Item Type	Drag and Drop (2 pts)
Stimulus	

Data Analysis

Item	State	Local
Full Credit	41	
No Credit	37	
Partial Credit	22	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.13(A) analyze the structures and functions of different species to identify how organisms survive in the same environment

2025 – Q14

Characteristics of the Atlantic puffin and the European starling are shown in the table.

Characteristic	Atlantic Puffin	European Starling
Habitat	Rocky coasts, open water of the North Atlantic Ocean	Cities, parks, open fields
Food Sources	Small fish, small shellfish	Insects, spiders, seeds, berries, fruit
Features	Wide, triangular beak; spiny tongue for holding prey; webbed feet; short wings	Long, pointed beak; strong legs and toes; long, triangular wings
Egg Laying	One egg laid per year	Four to six eggs laid twice per year
		

The European starling has a population size of about 150 million individuals, while the Atlantic puffin has a population size between 12 million and 14 million individuals.

Explain **TWO** likely reasons why the European starling has a greater population size than the Atlantic puffin.

Read the information in the table carefully. Then enter your answer in the box provided.

***Correct Answer (See Scoring Guide)**

Analysis of Assessed Standards

Cluster	Structure, Function, and Survival
Subcluster	Structure and Function
Content	Readiness
Process	
Item Type	Short Constructed Response (2 pts)
Stimulus	

Data Analysis

Item	State	Local
Full Credit	36	
No Credit	28	
Partial Credit	36	

Error Analysis

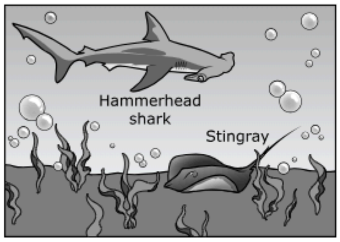
- ☐ Guessing
- ☐ Mixed Up Concepts
- ☐ Careless Error
- ☐ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.13(A) analyze the structures and functions of different species to identify how organisms survive in the same environment	**Analysis of Assessed Standards**

2025 – Q20

Students learned about ocean animals and made a table comparing some features of hammerhead sharks and stingrays.



Hammerhead Shark	Stingray
Sharp teeth	Flattened teeth
Long, tapered body with a very large head	Kite-shaped, flattened body
One eye on each side of its head	Eyes located on top of its head
Many large, low-density body parts	Few low-density body parts

Which piece of evidence provides the **LEAST** support for the claim that hammerhead sharks are adapted to live in open water and stingrays are adapted to live on the ocean floor?

Ⓐ A hammerhead shark has sharp teeth to bite and tear into its prey, while a stingray has flat teeth that can be used to crush prey.

Ⓑ A hammerhead shark's eyes are positioned in a way that allows it to see in many directions, while a stingray's eyes are positioned so that it can look upward.

Ⓒ The slender body shape of a hammerhead shark allows it to swiftly swim through the water, while the flat body shape of a stingray allows it to glide along the ocean bottom and settle into the sand.

Ⓓ A hammerhead shark has a body part that is less dense than water and prevents the shark from sinking, while the stingray does not have this body part that helps it float, so it will sink when it stops swimming.

***Correct Answer (A)**
^highly chosen incorrect answer (C)

Cluster	Structure, Function, and Survival
Subcluster	Structure and Function
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A*	32	
B		
C	^	
D		

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.13(A) analyze the structures and functions of different species to identify how organisms survive in the same environment

OLD **5.10(A)** compare the structures and functions of different species that help them live and survive in a specific environment such as hooves on prairie animals or webbed feet in aquatic animals

2024 – Q23

This question has two parts. First, answer Part A. Then answer Part B.

Characteristics of mosses and apple trees are shown in the table.

Mosses	Apple Trees
Grow on the surface of soil or on bare rock	Trunk and branches grow upward from roots in the soil
No roots; water absorbed by plants' outer surfaces	Water absorbed by roots and transported to leaves
No seeds; spores spread by wind and water	Seeds spread by animals
Life span of 2 to 10 years	Life span of 35 to 45 years
	

Part A
Which statement **BEST** describes a survival advantage apple trees could have compared to mosses in certain environments?

Ⓐ Apple trees have easier access to surface water.

Ⓑ Apples can be transported more easily by wind.

Ⓒ Apple trees have shorter life spans.

Ⓓ Apple seeds are eaten and moved to faraway locations by animals.

Part B
Which statement **BEST** supports the answer to Part A?

Ⓐ Wind easily moves apples to multiple locations away from the parent trees.

Ⓑ Shorter life spans lead to increased growth and development of the apples.

Ⓒ Apple seedlings will be less likely to compete with the parent tree for resources.

Ⓓ Competition for water is not likely to occur, since the tree uses the surface water before the moss does.

***Correct Answer (D, C)**

Analysis of Assessed Standards

Cluster	Structure, Function, and Survival
Subcluster	Structure and Function
Content	Readiness
Process	
Item Type	Multipart (2 pts)
Stimulus	

Data Analysis

Item	State	Local
Full Credit	20	
No Credit	67	
Partial Credit	13	

Error Analysis

- ☐ Guessing
- ☐ Mixed Up Concepts
- ☐ Careless Error
- ☐ Stopped Too Early

Learning from Mistakes Instructional Implications

5.13(A) analyze the structures and functions of different species to identify how organisms survive in the same environment

OLD **5.10(A)** compare the structures and functions of different species that help them live and survive in a specific environment such as hooves on prairie animals or webbed feet in aquatic animals

2023 – Q12

The table shows characteristics of two different types of birds that live in similar forest environments.

Great Hornbill	Macaw
	
• Long, curved beak • No bone in tongue • Long toes for grasping • Mostly black-colored body • Omnivore (insects, small animals, soft fruits)	• Short, curved beak • Tongue has bone • Long toes for grasping • Bright, colorful body • Herbivore (fruits, nuts, berries, seeds)

Which statement explains an advantage of a bone in the macaw's tongue that the hornbill does **NOT** need?

Ⓐ It helps macaws catch and chew small mammals and crustaceans.

Ⓑ It helps macaws break open and eat tropical fruits and nuts.

Ⓒ It helps macaws make large holes in trees to reach insects to eat.

Ⓓ It helps macaws swallow whole prey.

***Correct Answer (B)**

Analysis of Assessed Standards

Cluster	Structure, Function, and Survival
Subcluster	Structure and Function
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A	10	
B*	73	
C	9	
D	8	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.13(A) analyze the structures and functions of different species to identify how organisms survive in the same environment

OLD **5.10(A)** compare the structures and functions of different species that help them live and survive in a specific environment such as hooves on prairie animals or webbed feet in aquatic animals

2023 – Q19

The tallest trees in a rain forest are removed for lumber. This increases the amount of light on the forest floor. The table shows information about fungi and ferns that grow on the forest floor.

Wood-Decay Fungi	Fiddle Leaf Ferns
• Get energy from decaying wood • Make structures called spores to reproduce • Have spores that grow best in dark conditions • Use root-like structures to gather water	• Get energy from sunlight • Make structures called spores to reproduce • Have spores that need sunlight to grow • Have roots, stems, and leaves
 FreeImages.com/Jan Vermee	 Andrea Stickel/publicdomainpictures.net

Which statement explains why one organism is better adapted to the removal of the tall trees?

Ⓐ The fungi are better adapted because they have spores that grow better in the dark.

Ⓑ The ferns are better adapted because they have strong root systems.

Ⓒ The fungi are better adapted because they can get energy from decaying trees.

Ⓓ The ferns are better adapted because they get energy from the sun.

***Correct Answer (D)**

Analysis of Assessed Standards

Cluster	Structure, Function, and Survival
Subcluster	Structure and Function
Content	Readiness
Process	
Item Type	Multiple Choice (1 pt)
Stimulus	

Data Analysis

Item	State	Local
A	12	
B	13	
C	28	
D*	47	

Error Analysis

☐ Guessing ☐ Mixed Up Concepts

☐ Careless Error ☐ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.13(A) analyze the structures and functions of different species to identify how organisms survive in the same environment

OLD **5.10(A)** compare the structures and functions of different species that help them live and survive in a specific environment such as hooves on prairie animals or webbed feet in aquatic animals

Analysis of Assessed Standards

2022 – Q15

15 The foot of a hawk and the foot of a chicken are shown.

Hawk	Chicken
	

The difference between the shapes of their feet is most likely associated with the —

A predators that hunt them

B climate in which they live

C way they get their food

D distance they can fly

***Correct Answer (C)**

Cluster	Structure, Function, and Survival
Subcluster	Structure and Function
Content	Readiness
Process	5.2(D)
Stimulus	

Data Analysis

Item	State	Local
A	8	
B	13	
C*	77	
D	2	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

5.13(A) analyze the structures and functions of different species to identify how organisms survive in the same environment

OLD **5.10(A)** compare the structures and functions of different species that help them live and survive in a specific environment such as hooves on prairie animals or webbed feet in aquatic animals

Analysis of Assessed Standards

Cluster	Structure, Function, and Survival
Subcluster	Structure and Function
Content	Readiness
Process	5.2(D)
Stimulus	

Data Analysis

Item	State	Local
A	8	
B	21	
C*	67	
D	4	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes Instructional Implications

2022 – Q21

21 Butterflies have very long, tubelike tongues. Hummingbirds have very long tongues.

Which statement best describes why butterflies and hummingbirds both have long tongues?

A Butterflies and hummingbirds both migrate.

B Butterflies and hummingbirds are attracted to brightly colored flowers.

C Butterflies and hummingbirds eat the same food.

D Butterflies and hummingbirds have the same predators.

***Correct Answer (C)**

5.13(A) analyze the structures and functions of different species to identify how organisms survive in the same environment

OLD **5.10(A)** compare the structures and functions of different species that help them live and survive in a specific environment such as hooves on prairie animals or webbed feet in aquatic animals

Analysis of Assessed Standards

Cluster	Structure, Function, and Survival
Subcluster	Structure and Function
Content	Readiness
Process	5.2(D)
Stimulus	

2021 – Q35

35 The raccoon can be found in most areas of Texas. The coati has a smaller territory and is found from Big Bend to Brownsville. These two species are related. The chart shows the characteristics of these two species.

Raccoon

© iStock.com/GlobalP

- Sharp teeth and powerful claws
- Paws with flexible toes
- Good eyesight at night
- Eats fruits, berries, insects, rodents, frogs, fish, eggs, corn
- Long fluffy tail

Coati

© iStock.com/Oktay Ortakcioglu

- Sharp teeth and long claws
- Paws with flexible toes
- Good eyesight at night
- Eats insects, lizards, frogs, roots, fruits, nuts, and eggs
- Long tail that helps in balancing

Which statement does NOT describe how the body structures of these animals help them escape predators?

A Paws with long claws on flexible toes help them climb trees.

B Sharp teeth allow them to catch prey on land or in water.

C Long tails allow them to balance on branches in trees.

D Good eyesight helps them to see at night.

***Correct Answer (B)**

Data Analysis

Item	State	Local
A	**11**	
B*	**41**	
C	**35**	
D	**13**	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.13(A) analyze the structures and functions of different species to identify how organisms survive in the same environment

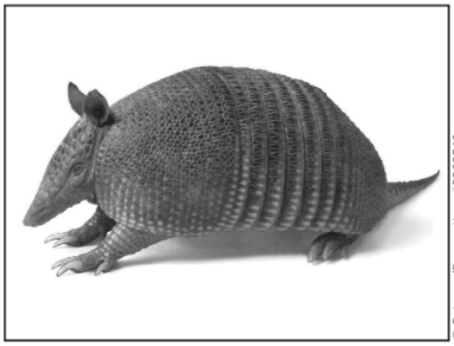
OLD **5.10(A)** compare the structures and functions of different species that help them live and survive in a specific environment such as hooves on prairie animals or webbed feet in aquatic animals

Analysis of Assessed Standards

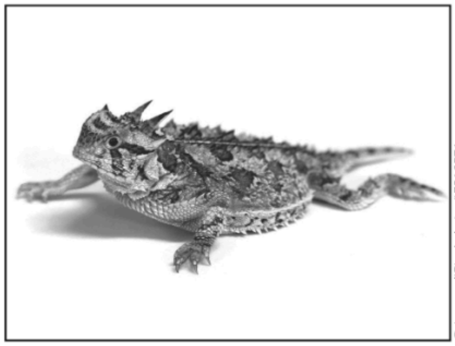
Cluster	Structure, Function, and Survival
Subcluster	Structure and Function
Content	Readiness
Process	5.2(D)
Stimulus	

2019 – Q26

The picture shows two animals that live in Texas.


© Batuque/Dreamstime #2260540

Texas Nine-banded Armadillo


© texcroc/iStockphoto #7216571

Texas Horned Lizard

Both animals have structures that help them survive. Which sentence best describes the function of the structures that these animals have in common?

F Both animals have small eyes that help them see clearly at night.

G Both animals have tough skin to keep them warm in cold weather.

H Both animals have sharp claws that help them to dig in sand.

J Both animals have pointed tails to attract predators.

***Correct Answer (H)**

Data Analysis

Item	State	Local
F	18	
G	19	
H*	58	
J	5	

Error Analysis

- ▢ Guessing
- ▢ Mixed Up Concepts
- ▢ Careless Error
- ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

5.13(A) analyze the structures and functions of different species to identify how organisms survive in the same environment

OLD **5.10(A)** compare the structures and functions of different species that help them live and survive in a specific environment such as hooves on prairie animals or webbed feet in aquatic animals

Analysis of Assessed Standards

Cluster	Structure, Function, and Survival
Subcluster	Structure and Function
Content	Readiness
Process	
Stimulus	

2019 – Q31

Unlike humans, otters have special flaps that close off their nostrils and ears. These flaps help otters survive in an environment that is —

A terrestrial

B snowy

C windy

D aquatic

Data Analysis

Item	State	Local
A	7	
B	10	
C	9	
D*	74	

Error Analysis

▢ Guessing ▢ Mixed Up Concepts

▢ Careless Error ▢ Stopped Too Early

Learning from Mistakes
Instructional Implications

***Correct Answer (D)**

IQ Analysis | Investigating the Question	SE	RC
Units		

Analysis of Assessed Standards

Cluster	
Subcluster	
Content	
Process	
Item Type	
Stimulus	

Data Analysis

Item	State	Local

Error Analysis

- ☐ Guessing
- ☐ Careless Error
- ☐ Stopped Too Early
- ☐ Mixed Up Concepts

Learning from Mistakes
Instructional Implications

*** Correct Answer ( )**

Analysis of Assessed Standards

Cluster	
Subcluster	
Content	
Process	
Item Type	
Stimulus	

Data Analysis

Item	State	Local

Error Analysis

- ☐ Guessing
- ☐ Careless Error
- ☐ Stopped Too Early
- ☐ Mixed Up Concepts

Learning from Mistakes
Instructional Implications

*** Correct Answer ( )**